

Solving equations by radicals

A mathematical drama on Galois theory

Hao Chen

*Department of Electrical and Computer Engineering, Princeton University,
Princeton, NJ 08540 USA*

*Department of Physics, Princeton University,
Princeton, NJ 08540 USA*

E-mail: chen.hao@princeton.edu

ABSTRACT: This manuscript provides a concise yet concrete narrative – or “drama” – of the Galois theory’s approach to the solvability of polynomial equations by radicals, culminating in the proof of the non-existence of quintic formulas. Based on a series of online lectures delivered in February 2026, the text assumes only a high-school level of mathematical maturity. It offers a self-contained introduction to field extensions and the core tenets of Galois theory, centering strictly on the fundamental problem of radical solvability. Having been successfully tested with general audiences, this material is well-suited for students and non-specialists seeking an accessible entry point into abstract algebra.

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1 Preface

This manuscript aims to give a concise but concrete lecture (maybe “drama” is more suitable) on the Galois theory approach to the solvability of polynomial equations by radicals, paving a way to the non-existence of quintic formula – a classic topic in algebra. The main goal is to present the key ideas and results in a way that is accessible to non-specialists, while still being self-contained and mathematically rigorous.

We find this topic to be particularly suitable for non-specialists to get a taste of abstract algebra and to appreciate the power of algebraic structures in dealing with concrete problems. The content of the drama is designed in a question-oriented manner, starting from the basic question of solving polynomial equations by radicals and guiding the audience through the necessary concepts like field extensions and Galois groups via follow-up questions that naturally arise.

To keep the drama concise, attractive, and easy to follow, we have tried to avoid any concept, theorem, or proof that is not directly relevant to the main question, even though they play important roles in the theory of algebra. Some technical details and proofs which may distract the audience from the mainstream but are essential for the drama to be self-contained are deferred to the appendix for interested readers.

However, even though the drama is logically self-contained, it is quite common that a wide range of audience members may find some parts hard to follow and would like to know additional details. To address this issue, we refer readers to a great introductory-level textbook: *A Book of Abstract Algebra* by Charles C. Pinter [1]. For many foundational algebraic topics that play supporting roles in our drama, we have provided references to the corresponding pages in Pinter’s book. For example, “xxxx^{P110–115}” means the relevant details can be found on pages 110–115 of Pinter’s book.

Along our way to the non-existence of quintic formulas, we will make a few claims that may need some thoughts to justify, but the drama would look too gross if we listed all of them as theorems and provided rigorous proofs. Instead, we will mark them with “[exercise]” for the audience to verify by themselves. We promise that all these verifications can be done by straightforward arguments or calculations.

Nowadays, generative AI tools are dramatically changing our way of learning. When designing this drama, we have checked that every “[exercise]” can be smoothly solved by the AI agents built upon the latest large language models¹. The audience are encouraged to consult AI agents for filling in any gaps in their understanding while watching the drama.

We hope the audience will find this drama helpful and inspiring!

¹Examples include GPT-5.2 and Gemini 3. One may need to use “thinking” mode (the AI agent generates an internal monologue or chain-of-thought to enhance reasoning.) for best performance.

2 The question and basic concepts

2.1 The question: solving equations by radicals

Is it possible to solve a polynomial equation $a(x) = 0$ by radicals? That is, to express all its solutions (roots) by radical expressions of its coefficients. More precisely:

- Expressions built up by repeated addition, subtraction, multiplication, division, and taking roots are called **radical expressions**.
- A polynomial $a(x)$ is **solvable** (by radicals) if there is a radical expression giving (all of) its roots in terms of its coefficients².

Example 2.1. As a random example, $\frac{\sqrt{7}x^2+8x^3}{\sqrt[3]{5+\sqrt{3}x}}$ can be seen as a radical expression of $x, 7, 8, 5, 3$, or a radical expression of $x, 5, \sqrt{3}, \sqrt{7}, 2$ (as $8 = 2 * 2 * 2$).

2.2 A subtlety: solvability vs. generic formula for roots

Wait! We have to emphasize something before proceeding.

We understand that the audience of this drama comes with the curiosity of how the young variste Galois was able to show the non-existence of quintic formulas, but there is a gap between that and the solvability we defined in above: quintic formulas are *generic* radical formulas, taking the coefficients as *variables*, that can express *all* the solutions of *any* quintic equation, while solvability is about expressing the solutions of a *specific* polynomial equation $a(x) = 0$ using radicals involving the coefficients of this *specific* polynomial and some other numbers (to be elaborated: see Sec. 2.6).

So here is the idea: if there are generic radical formulas to express all the solutions of every equation in a set, then each of these equations must be solvable by radicals; while even if there is one equation in the set that is not solvable by radicals, we can conclude that no generic radical formulas for roots exist.

2.3 Galois's approach: a theory of solvability for every equation

The approach by Évariste Galois to the non-existence of quintic formula is as follows: he established the profound theory named after himself that gives the *necessary and sufficient* condition for solvability of polynomial equations. Then finding one quintic equation that is not solvable would be enough to exclude the existence of quintic formulas.

Galois theory apparently provides much more than needed for showing non-existence of quintic formulas. Moreover, Évariste Galois was not one of the first people to arrive at this conclusion; rather, it is called the **Abel-Ruffini theorem**, named after Paolo Ruffini, who made an incomplete proof in 1799 (which was refined and completed in 1813 and

²This is to be elaborated, as the radicals are allowed to contain some numbers other than the coefficients.

accepted by Cauchy), and Niels Henrik Abel, who provided a proof in 1824.

Nevertheless, Galois theory dramatically supersedes the work of Abel and Ruffini (Allan Clark, *Elements of Abstract Algebra* [2]) as it provides a deep connection between field theory and group theory that allows reducing certain problems in field theory to group theory.

2.4 Basic concepts: Polynomials and fields

Definition 2.1 (Polynomial over a field). *Given a field F and an arbitrary symbol x , the expression*

$$a(x) \equiv a_0 + a_1x + a_2x^2 + \cdots + a_nx^n, \quad (2.1)$$

*where $a_k \in F$ for $k \in 1, \dots, n$, is called a **polynomial in x over F** .*

Here, every a_kx^k is called a **term** of the polynomial $a(x)$ and a_k is called the **coefficient** of x^k . F is called the **coefficient field**. The greatest n such that $a_n \neq 0$ is the **degree** of $a(x)$, denoted as $\deg a(x) = n$. The set of all polynomials over F is denoted by $F[x]$.

A possibly unfamiliar concept is **field**³. Briefly speaking, a field is a set on which **arithmetic operations** $+$, $-$, $\times$, $\div$ are defined⁴ and behave like the corresponding operations on rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$, and complex numbers $\mathbb{C}$.

2.5 Why fields matter

Galois theory's approach to the solvability of polynomial equations is closely tied to the *fields* that contain their coefficients and solutions. Here is the key insight:

By a radical expression involving elements in a field F , what numbers can be reached?

- **Addition (+), subtraction (-), multiplication ($\times$), division ($\div$):**

By the definition of a field, for all elements in a field F , any expression built from the arithmetic operations stays in F .

- **Root-taking:**

The result can *escape* the original field F !

(Example: $\sqrt{2}$ is not rational, even though 2 is.)

³We will only be concerned with fields of numbers with ordinary $+$, $-$, $\times$, $\div$. See the appendix for rigorous definitions of field (see A.17) and other algebraic concepts. $F[x]$ is usually not a field but an integral domain (see A.16).

⁴In algebra, an operation is a map which takes two elements in the set as its input and yields one element as the output. An operation is said to be **defined on a set** if any two elements in the set can be taken as input for the operation to give a unique output. We also say that the set is **closed** under an operation defined on it.

So it is fine if the roots of $a(x)$ are not already in its coefficient field F : since the radicals can go out of F , extending the realm of numbers we can reach, we can still possibly reach all the roots of $a(x)$.

2.6 Restating solvability with the language of fields

Now we can elaborate the missing part of solvability:

Definition 2.2 (Solvability of a polynomial). *A polynomial $a(x)$ over F is **solvable by radicals** if all its roots can be expressed by radical expressions over F , i.e., **radicals in terms of elements in its coefficient field F** .*

This alerts us that one can only talk about whether an expression is radical *with respect to a field*; therefore, the rigorous meaning of quintic formulas – “generic radical formulas for roots” – also needs to be updated accordingly: a set of formulas that can express all the roots of every quintic polynomial in terms of radicals over the coefficient field of that polynomial, taking the coefficients as variables.

Since having one quintic polynomial being not solvable by radicals is enough to exclude the existence of quintic formulas, we will from now on focus on investigating solvability, as Galois did.

2.7 Approaching roots by extending coefficient field

Intuitively, if we take a large enough field K , the following can happen:

Definition 2.3 (Splitting of a polynomial). *A polynomial $a(x)$ over F is said to **split** in a field K if it decomposes into linear factors, i.e., if*

$$a(x) = a_n(x - r_1)(x - r_2) \dots (x - r_n), \quad (2.2)$$

where $r_k \in K$ are the n roots of $a(x)$.

Example 2.2. $a(x) = x^2 - 2 = (x + \sqrt{2})(x - \sqrt{2})$ does not split in $\mathbb{Q}$ since its roots $\pm\sqrt{2} \notin \mathbb{Q}$, but it splits in a larger field, e.g., $\mathbb{R}$, which contains all its roots $\pm\sqrt{2}$.

With the above idea in mind, the notion of solvability can be reformulated in a more field theoretic way:

The crux of solvability: A polynomial is solvable by radicals if and only if, through a finite⁵ sequence of root-taking operations, we can *extend* the coefficient field F step-by-step to a large enough field containing *all the roots*.

The next section is all about extending fields.

⁵It is reasonable to require finite amount of operations, in accordance with the traditional sense of “solving equations”. Moreover, it is critical because allowing infinite operations opens a Pandora’s box: infinite arithmetic operations can express basically anything, e.g., $\sqrt{2} = 1 + \frac{1}{2 + \frac{1}{2 + \dots}}$, trivializing the notion of solvability.

3 From equations to field extensions

3.1 Field extensions: larger fields containing new numbers

Starting from a field F , we want to create a larger field K that contains F and additional new elements.

Definition 3.1 (Field extension). A **field extension** $K : F$ (read as “ K over F ”) is a pair of fields where $F \subseteq K$, i.e., F is a **subfield**⁶ of K . We say K is an **extension field** of F ⁷.

What does extending mean? We take a field F and some new elements $c_1, c_2, \dots, c_n \notin F$ to create a larger field K that:

- contains all elements of F and all the new elements $c_1, c_2, \dots, c_n$;
- is still a field, i.e., still closed under addition, subtraction, multiplication, and division.

It is easy to perceive that such K is not unique (e.g., both $\mathbb{R}$ and $\mathbb{C}$ contain $\mathbb{Q}$ and $\sqrt{2}$), and for K to be a field, it typically contains many more elements than just F and $c_1, c_2, \dots, c_n$.

3.2 Algebraic and transcendental elements

Before proceeding, let's take a brief digression:

Definition 3.2 (Algebraic and transcendental elements). Let $F \subseteq K$. An element $c \in K$ is said to be **algebraic** over F if there exists a non-zero polynomial $a(x) \in F[x]$ such that $a(c) = 0$, i.e., c is a root of some polynomial $a(x)$ with coefficients in F . Otherwise, c is called **transcendental** over F .

For instance, $\sqrt{2}$, a root of $x^2 - 2 = 0$, is algebraic over $\mathbb{Q}$, while π is transcendental over $\mathbb{Q}$. Proving an arbitrary number to be transcendental is a non-trivial task (e.g., the transcendence of π and e) and beyond the scope of algebra (usually involves analysis).

Fortunately, both radical expressions of coefficients and roots of polynomials are always algebraic over the coefficient field [exercise]. All elements in the following discussions are algebraic over the base field unless otherwise specified.

3.3 Simple extension: adding one new element to a field

As field extensions of a given field and new elements are not unique, a natural question arises: Given a field F and one element $c \notin F$, what is the *smallest* field that contains both F and c ?

⁶A subfield is, as you expect, any nonempty subset of the original field which is closed with respect to the $+$, $-$, $\times$, $\div$ of the original field.

⁷Strictly speaking, a field extension (like $K : F$) is totally different from an extension field (K itself). However, the two words are often used interchangeably in literature, so we will adopt this convention when there is no risk of causing confusions.

Definition 3.3 (Simple extension). *The smallest field containing both F and one new element $c \notin F$ is denoted by $F(c)$, called a **simple extension field** of F obtained by adjoining c to F . The extension $F(c) : F$ is then called a **simple extension** over F .*

It can be shown^{P272–274} that for c being algebraic over F , any field containing F and c would inevitably contain every element of the form

$$a_0 + a_1c + \cdots + a_m c^m \quad (a_0, \dots, a_m \in F, m \in \mathbb{Z}^{0+}). \quad (3.1)$$

The set of all such elements forms a field, which is $F(c)$.

It requires learning quite a bit of abstract algebra to prove the above result, especially what is so special about c being algebraic such that the multiplicative inverse of an element like (3.1) also has the same form. (This is crucial for the set of all such elements to be closed under division.) However, to keep you focus on the mainstream, we sweep the details under the rug. We apologize for this and promise you that this is the only point in this drama that is not self-contained. For your satisfaction, you can take “the set containing all elements of the form (3.1)” as the definition of $F(c)$, while try convincing yourself that it really is a field (closed under arithmetic operations) through the following examples.

Example 3.1. *What is the structure of $\mathbb{Q}(\sqrt{2})$, the smallest field obtained by adjoining $\sqrt{2}$ to the rationals $\mathbb{Q}$?*

Answer: According to (3.1), every element of $\mathbb{Q}(\sqrt{2})$ can be written uniquely as

$$a + b\sqrt{2} \quad \text{where } a, b \in \mathbb{Q}.$$

Sanity check: $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is indeed a field:

- All rational numbers ($b = 0$) and $\sqrt{2}$ ($a = 0, b = 1$) is in the field.
- Addition and subtraction: $(a + b\sqrt{2}) \pm (c + d\sqrt{2}) = (a \pm c) + (b \pm d)\sqrt{2}$ also has the form.
- Multiplication: $(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$ also has the form.
- For division: $\frac{a+b\sqrt{2}}{c+d\sqrt{2}} = \frac{(a+b\sqrt{2})(c-d\sqrt{2})}{c^2-2d^2} = \frac{(ac-2bd)+(bc-ad)\sqrt{2}}{c^2-2d^2}$ also has the form $a' + b'\sqrt{2}$.

The only nontrivial result from above is that $\{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is closed under division. We managed to show it by rationalizing the denominator.

Example 3.2. *What is the structure of $\mathbb{Q}(\sqrt[3]{2})$?*

Answer: According to (3.1), every element of $\mathbb{Q}(\sqrt[3]{2})$ can be written uniquely as

$$a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2 \quad \text{where } a, b, c \in \mathbb{Q}.$$

One can verify [exercise] that $\mathbb{Q}(\sqrt[3]{2})$ is indeed a field by checking closure under $+, -, \times, \div$ as in the previous example⁸.

It is now evident that $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt[3]{2})$ have different “structures”, even though both are obtained by adjoining a single element to $\mathbb{Q}$. As a comparison, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ have the *same* “structure”, as both can be expressed as $a + b\sqrt{d}$ for $d = 2$ or 3 .

This so-called “structure” of a field extension is captured by its *degree*, which we will discuss next.

3.4 Degree of field extensions

Key idea: Given a field extension $K : F$, we can view K as a *vector space* over F . The **degree** of the extension, denoted $[K : F]$, is the dimension of this vector space. We say K is an *extension of degree n over F* .

Recall what is a vector space^{P282–291}: If $K : F$ has degree n , then there exist n elements $v_1, v_2, \dots, v_n$ in K such that:

- The set $\{v_1, v_2, \dots, v_n\}$, called a **basis**, is linearly independent over F , i.e., the only set of scalars $a_1, a_2, \dots, a_n \in F$ such that $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$ are all $a_k = 0$.
- Every element of K can be expressed $a_1v_1 + a_2v_2 + \dots + a_nv_n$, called a **linear combination** of some $a_k \in F$.

Example 3.3. In $\mathbb{Q}(\sqrt{2}) : \mathbb{Q}$ the basis is $\{1, \sqrt{2}\}$, as every element can be expressed as $a \cdot 1 + b \cdot \sqrt{2}$ with $a, b \in \mathbb{Q}$. Its degree is $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$.

Example 3.4. In $\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}$ the basis is $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$, as every element can be expressed as $a \cdot 1 + b \cdot \sqrt[3]{2} + c \cdot (\sqrt[3]{2})^2$ with $a, b, c \in \mathbb{Q}$. Its degree is $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$.

3.5 The degree and a basis of $F(c) : F$

Given a field F and an element $c \notin F$ that is algebraic over F , what is the degree of the extension $F(c) : F$? As mentioned in (3.1), every element of $F(c)$ can be expressed as a polynomial in c with coefficients from F :

$$a_0 + a_1c + a_2c^2 + \dots + a_mc^m$$

⁸Verifying $\mathbb{Q}(\sqrt[3]{2})$ is closed under division involves rationalizing the denominator in a more tricky way. However, to find the inverse of $a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2$ we can always assume it is $p + q\sqrt[3]{2} + m(\sqrt[3]{2})^2$. Their product is one so we have $(ap + 2cq + 2bm) + (bp + aq + 2cm)\sqrt[3]{2} + (cp + bq + am)(\sqrt[3]{2})^2 = 1$, that is

$$\begin{cases} ap + 2cq + 2bm = 1 \\ bp + aq + 2cm = 0 \\ cp + bq + am = 0 \end{cases} \quad (3.2)$$

This set of linear equations of (p, q, m) actually always has a unique solution for rational a, b, c , unless $a = b = c = 0$, though proving this result by directly showing the coefficient matrix has nonzero determinant requires some number theory that we will skip here.

where $a_k \in F$ and $m \in \mathbb{Z}^{0+}$. To know $[F(c) : F]$, we need to know how many of these elements are *linearly independent* over F . Just as in $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$, we found that 1 and $\sqrt{2}$ can be considered linearly independent over $\mathbb{Q}$ and span $\mathbb{Q}(\sqrt{2})$, giving $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. The following concept is the key:

Definition 3.4 (Minimal polynomial). *The **minimal polynomial** of c over F is the monic polynomial⁹ $p(x) \in F[x]$ of lowest degree having c as a root: $p(c) = 0$.¹⁰*

Theorem 3.1 (The degree of $F(c) : F$). *Using the division algorithm^{P245–246}, it can be shown^{P293} that if c is algebraic over F with minimal polynomial $p(x)$ of degree n , then every element of $F(c)$ can be expressed uniquely as a linear combination:*

$$a_0 + a_1c + a_2c^2 + \cdots + a_{n-1}c^{n-1} \quad (a_k \in F), \quad (3.3)$$

and from which one can prove [exercise] that the set $\{1, c, c^2, \dots, c^{n-1}\}$ forms a basis of $F(c)$ over F , giving rise to the degree: $[F(c) : F] = n$.

It may seem like we've been spending too much time on the simple case of adjoining only one element, but it is indeed a crucial building block for understanding more complex extensions formed by adjoining multiple elements step-by-step, which is, as we will see soon, exactly what we do when constructing field extensions by radicals (root-taking).

3.6 Iterated extensions: adjoining multiple elements

Definition 3.5 (Iterated extensions). *If we want to adjoin multiple elements $c_1, c_2, \dots, c_m$ to a field F , we can do so step-by-step:*

$$F \subseteq F(c_1) \subseteq F(c_1, c_2) \subseteq \cdots \subseteq F(c_1, c_2, \dots, c_m).$$

*In the above **extension tower**, each step is a simple extension, i.e., $F(c_1, c_2, \dots, c_i)$ is the smallest field containing $F(c_1, c_2, \dots, c_{i-1})$ and c_i . Such an extension $F(c_1, \dots, c_m) : F$ is called an **iterated extension**.*

An immediate concern is: **does the order of adjoining elements matter?** The answer is **no**: by the requirements of a field (closed under arithmetic operations), every field containing $F(c_1)$ and c_2 must also contain F and both c_1 and c_2 , and vice versa. Thus, $F(c_1, c_2)$ is the smallest field containing F and both c_1 and c_2 , making the order of adjoining irrelevant.

3.7 Degree and basis of iterated extensions

Theorem 3.2 (Degree of extension tower). *Suppose $F \subseteq K \subseteq E$ where $[E : K]$ and $[K : F]$ are both finite (field extensions with finite degrees are called **finite extensions**). Then $E : F$ is also a finite extension such that*

$$[E : F] = [E : K][K : F]. \quad (3.4)$$

⁹A monic polynomial is one whose leading coefficient is 1.

¹⁰Sometimes, we need to loosen the requirement of “monic” to talk about the minimal polynomial $p(x)$ multiplied by some constant $b \in F$. Since $bp(x)$ is also a polynomial of the lowest degree having c as a root, we call it “a minimal polynomial”, whereas the special title “the minimal polynomial” is reserved for the monic $p(x)$.

This is a powerful tool in the study of fields, analogous to Lagrange's theorem^{P129} in group theory. The proof of it is based on a fact from linear algebra:

Theorem 3.3 (Product basis of vector spaces). *Let $\{a_1, \dots, a_m\}$ be a basis of the vector space K over F , and let $\{b_1, \dots, b_n\}$ be a basis of the vector space E over K . Then the set of mn products $\{a_i b_j\}$ is a basis of the vector space E over the field F .*

The proof of this fact can be found in the appendix [A.18](#).

Based on the general theorems above, the degree of an iterated extension is:

$$[F(c_1, \dots, c_m) : F] = [F(c_1, \dots, c_m) : F(c_1, \dots, c_{m-1})] \times \cdots \times [F(c_1) : F].$$

A basis of $F(c_1, \dots, c_m)$ over F can be formed by taking all possible products of basis elements from each simple extension step, that is, terms of the form

$$c_1^{i_1} c_2^{i_2} \cdots c_m^{i_m}, \tag{3.5}$$

where the exponents i_k run from 0 to one less than the degree of the simple extension at each step.

Practically, it is hard to keep track of the exact ranges of every i_k , but we only need to keep in mind that every linear combination of $c_1^{i_1} c_2^{i_2} \cdots c_m^{i_m}$ with coefficients from F and exponents $i_k \in \mathbb{Z}^{0+}$ is an element of $F(c_1, \dots, c_m)$.

Example 3.5. *What is the structure of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$?*

Answer: *The distinct terms we can possibly get by multiplying a couple of $\sqrt{2}$ and $\sqrt{3}$ together are $\sqrt{2}, \sqrt{3}, \sqrt{6}$, then according to (3.5), every element of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ can be uniquely written as $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$, where $a, b, c, d \in \mathbb{Q}$, from which the degree of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is clearly 4.*

You can do a sanity check by showing the $+, -, \times, \div$ of any two such numbers are still in this form, but we will practice using (3.4) here. Let's say we first adjoin $\sqrt{2}$ to $\mathbb{Q}$ and then adjoin $\sqrt{3}$. There is no question that $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$, but what is $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})]$? Evidently, $\sqrt{3}$ is not in $\mathbb{Q}(\sqrt{2})$ [exercise], so it is not a root of a polynomial of degree 1 over $\mathbb{Q}(\sqrt{2})$ (which would have to be $x - \sqrt{3}$). But $\sqrt{3}$ is a root of $x^2 - 3$, a polynomial of degree 2, serving as the minimal polynomial of $\sqrt{3}$ over $\mathbb{Q}(\sqrt{2})$; therefore, $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] = 2$. All together, we have $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2 \cdot 2 = 4$.

3.8 More facts about finite extensions

Before concluding the general discussion on field extensions, we point out two more important facts about finite extensions:

1. From the above discussions, we can deduce that **if $c_1, \dots, c_m$ are algebraic over F , then $F(c_1, \dots, c_m)$ is a finite extension of F** , because the degree of each step is finite. Actually, the converse is also true: **every finite extension is an iterated extension.**

This is obvious because a finite extension field K of F with degree n is a vector space over F with dimension n . Then, K has a basis $\{a_1, \dots, a_n\}$ over F , giving rise to the fact that K is the smallest field containing F and all a_i : $K = F(a_1, \dots, a_n)$.

2. If $K : F$ is a finite extension, then every element of K is algebraic over F .

Indeed, for any $c \in K$, the set $\{1, c, c^2, \dots, c^n\}$ with $n = [K : F]$ must be linearly dependent over F (as there are $n + 1$ elements in an n -dimensional vector space).

Thus, there exist $a_0, a_1, \dots, a_n \in F$, not all zero, such that $a_0 + a_1c + a_2c^2 + \dots + a_nc^n = 0$, i.e. c is a root of this polynomial.

3.9 Revisiting the main question: how field extensions help to understand solvability of polynomial equations?

Now we are equipped with the necessary knowledge of field extensions to get back to our main question about solving polynomial equations by radicals. As a reminder, Galois theory's approach to this question is based on the idea of extending from the coefficient field F by adjoining radical expressions step-by-step. If we can finally reach a field containing all the roots of the polynomial, then the polynomial is solvable by radicals.

Then, we are confronted with these crucial questions:

1. What are the fields we can reach from the coefficient field F by adjoining radical expressions step-by-step?
2. What is the (smallest) field containing all the roots of a given polynomial?
3. How to know if the first fields could contain the second field or not?

We will address the first two questions now, while the third question is answered by the protagonist of this drama: the celebrated Galois theory.

3.10 Radical extensions: building larger fields by taking roots

Definition 3.6 (Radical extension). *A finite extension $F(c_1, \dots, c_n)$ is called a **radical extension** of F if, for each k , some (integral) power of c_k lies in the previous field $F(c_1, \dots, c_{k-1})$. In other words, there exist positive integers $i_1, i_2, \dots, i_n$ such that for each k ,*

$$c_k^{i_k} \in F(c_1, c_2, \dots, c_{k-1}). \quad (3.6)$$

It means that each c_k can be obtained by taking an i_k -th root of some element in the previous field.

Then it is evident that, by radical expressions of elements in the coefficient field F , the numbers we can reach are precisely those in some radical extension of F .

Therefore, the polynomial $a(x) \in F[x]$ is solvable if and only if there exists a radical extension of F that contains all the roots of $a(x)$.

3.11 The root field: the smallest field containing all roots

It turns out that investigating whether a radical extension includes a field containing all the roots is simpler than directly checking if it contains the roots themselves, since the field structure provides richer information than just the individual roots. Therefore, the following concept naturally arises:

Definition 3.7 (Root field). *The **root field** of a polynomial $a(x)$ of degree n over F is the smallest extension field of F containing all n roots $r_1, \dots, r_n$ of $a(x)$. Clearly, the root field is the finite extension $F(r_1, \dots, r_n)$ ¹¹.*

In the literature, the root field is often called the **splitting field** of the polynomial, as the polynomial “splits” into linear factors over this field: $a(x) = a_n(x - r_1)(x - r_2) \cdots (x - r_n)$, where $a_n \in F$.

The criterion for solvability now translates into: There exists a radical extension of F that contains the root field $F(r_1, \dots, r_n)$.

3.12 [Optional]: existence of n roots

We’ve mentioned several times that a polynomial of degree n has n roots. It may sound very familiar because:

Theorem 3.4 (Fundamental Theorem of Algebra). *Every non-constant polynomial $a(x)$ with complex coefficients: $a(x) \in \mathbb{C}[x]$ has at least one complex root $r_1 \in \mathbb{C}$.*

Consequently, we can factor $a(x)$ as $a(x) = (x - r_1)b(x)$ where $b(x)$ is a polynomial of degree $n - 1$. Repeating this process, we show there are n roots (counting multiplicities) in $\mathbb{C}$. Although the proof relies on complex analysis and many people haven’t seen it before, almost everyone takes it for granted since high school.

However, this is not actually what we want here, as our coefficient field F should not be taken as $\mathbb{C}$, because otherwise, F already contains all the roots and there is no need to do field extensions at all!

In other words, **if we take the coefficient field F to be $\mathbb{C}$ in the first place, every polynomial in $F[x]$ is solvable by radicals since all roots are already in F .** Then by this approach, the solvability becomes trivial and can tell us nothing about the non-existence of quintic formula.

So how can we know there are n roots in the first place, provided that the only field F we have in hand may contain none of them? Here’s the saver:

Theorem 3.5 (Basic theorem of field extensions^{P275–276}). *Let F be a field and $a(x)$ a nonconstant polynomial of degree n in $F[x]$. There exists an extension field K of F and a $r_1 \in K$ such that $a(r_1) = 0$.*

¹¹If $r_k \in F(r_1, \dots, r_{k-1})$, we understand $F(r_1, \dots, r_k) = F(r_1, \dots, r_{k-1})$.

This is to say, in field K , $a(x)$ can be factored into $a(x) = (x - r_1)b(x)$, and $b(x) \in K[x]$ of degree $n - 1$. Repeating this process, we show that there exists an extension field E of F that contains n roots of $a(x)$, in which $a(x)$ splits into linear factors: $a(x) = a_n(x - r_1)(x - r_2) \cdots (x - r_n)$.

4 Intermission: getting a taste of Galois theory by “trivial” examples

4.1 Intermission: Why pausing here?

We hope the audience enjoyed the first half of this drama, gaining a basic understanding of field extensions and their relevance to expressing the roots of polynomial equations using radicals.

We understand that you may be eager to see how Évariste Galois was able to build upon these concepts to establish his profound theory and show the limitations of radicals for expressing the roots of polynomials of degree five and higher. However, we urge you to take a moment to pause here before diving into the real drama of Galois theory, and instead reflect on the concepts introduced so far by taking a close look at some equations with evident radical solutions.

You may scoff at the equations we present, as their solutions are too obvious, but what we have just learned allows us establish better understandings of why their solutions can be expressed by radicals, and more importantly, how the field extensions and their structures play a crucial role in this process.

4.2 Quadratic equations: the root field

[In the entire intermission, we think of the coefficient field of every polynomial to be $\mathbb{Q}$ unless otherwise noted.]

We all like the quadratic formula for solving equation $a(x) = ax^2 + bx + c = 0$ we learned in (junior) high school:

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

As $\Delta = b^2 - 4ac$ is not always a complete square, it suggests that for some quadratic equations, e.g., $x^2 - 2 = 0$, $\sqrt{\Delta}$ is not in its coefficient field, hence nor the roots $r_{1,2}$.

In such situations, the root field of $a(x)$ is $K = \mathbb{Q}(\sqrt{\Delta})$ [exercise], a radical extension of the coefficient field $\mathbb{Q}$. By (3.3), every element in K can be uniquely expressed as

$$p + q\sqrt{\Delta}, \quad p, q \in \mathbb{Q}.$$

4.3 Quadratic equations: an interesting observation

Now we want to draw everybody’s attention to an interesting observation. First of all, we can rewrite every element in K in the following two ways:

$$\begin{aligned} p + q\sqrt{\Delta} &= (p + bq) + (2aq)\frac{-b + \sqrt{\Delta}}{2a} = p_1 + q_1r_1, \quad p_1, q_1 \in \mathbb{Q}, \\ p + q\sqrt{\Delta} &= (p - bq) - (2aq)\frac{-b - \sqrt{\Delta}}{2a} = p_2 + q_2r_2, \quad p_2, q_2 \in \mathbb{Q}, \end{aligned}$$

suggesting that r_1 is on an equal footing with r_2 in K ¹².

To see it more clearly, consider a mapping $\sigma : K \rightarrow K$ that sends every element $x = p + qr_1$ to another element $\sigma(x) = p + qr_2$ in K , that is, replacing the role of r_1 with that of r_2 (specifically, $\sigma(r_1) = r_2$ and $\sigma(r_2) = r_1$ [exercise, see hints below]). Let's see what is special about this mapping.

Above all, $\sigma(x) = x, \forall x \in F$ – it leaves the coefficient field F unchanged.

With $r_1 + r_2 = \frac{-b}{2a}$, we realize $p + qr_2 = p + q(-r_1 - \frac{-b}{2a})$. So sticking to the basis $\{1, r_1\}$, we see the mapping σ is actually linear:

$$\sigma(p + qr_1) = (p + qb/2a) - qr_1.$$

A linear map by definition respects addition and subtraction relations: $\sigma(x \pm y) = \sigma(x) \pm \sigma(y)$. Actually, σ also respects multiplication $\sigma(x \cdot y) = \sigma(x) \cdot \sigma(y)$ and division $\sigma(x^{-1}) = [\sigma(x)]^{-1}$ [exercise; see appendix A.19]. Thus σ respects all field operations. Moreover, this map σ is bijective¹³, that is, a *one-to-one correspondence* between every element in the domain and every element in the codomain.

All these properties give rise to several critical concepts in field theory that are crucial for us to understand Galois theory, as we will discuss next.

4.4 Field isomorphism and automorphism (self-isomorphism)

Let's first focus on respecting field operations.

Definition 4.1 (Field isomorphism). A bijective map $\sigma : F_1 \rightarrow F_2$ such that for all elements $x, y \in F_1$:

$$\sigma(x + y) = \sigma(x) + \sigma(y), \quad \sigma(x \cdot y) = \sigma(x) \cdot \sigma(y)$$

is called an **isomorphism** from field F_1 to field F_2 .

With isomorphism σ respecting addition and multiplication, it follows that $\sigma(0) = 0$, $\sigma(1) = 1$, $\sigma(-x) = -\sigma(x)$, $\sigma(x^{-1}) = [\sigma(x)]^{-1}$ [exercise, see appendix A.20]. In short, a field isomorphism is a bijective map respecting all field operations (the four arithmetic operations $+, -, \times, \div$).

Intuitively, we think of a field isomorphism as a *one-to-one* correspondence of elements in one field with elements in another field that carries over all the rules of $+, -, \times, \div$. The mapping $\sigma(p + qr_1) = p + qr_2$ we discussed before is an isomorphism from the root field K to *itself*. Such an isomorphism from a field to itself is given the special name: **automorphism**.

¹²It is not hard to verify that $K = \mathbb{Q}(r_1) = \mathbb{Q}(r_2)$.

¹³Indeed, $\forall x \in K, \sigma(\sigma(x)) = x$, i.e., σ is an involution which is always bijective.

4.5 F -fixing automorphism as symmetry of the root field

Recalling that K is an extension field of the coefficient field F , and the mapping $\sigma(p+qr_1) = p+qr_2$ is an automorphism of K that leaves F unchanged, we call it an **F -fixing automorphism of the root field K** .

In mathematics and natural science, we use the term **symmetry** to describe a map (sometimes called a transformation) acting on an object that leaves something unchanged¹⁴. Then, an F -fixing automorphism of the root field K can be seen as a *symmetry* of K leaving the coefficient field F inside unchanged.

Since two successive F -fixing automorphisms still respect field operations and leave F unchanged, it is evident that all the F -fixing automorphisms of K form a *group*¹⁵ G such that the group operation of two automorphisms $\alpha, \beta \in G$ is the composition $\alpha \circ \beta$. Indeed, a group is a suitable algebraic structure to study symmetries of various stuff in the real world.

4.6 Quadratic equations: summary

This is the end of our exploration on quadratic equations and their root fields. Let's summarize what we have found to avoid possible confusions:

For a quadratic equation $ax^2 + bx + c = 0$ over $\mathbb{Q}$ with two roots $r_{1,2}$, if $\sqrt{\Delta} \notin \mathbb{Q}$, its root field is an extension field $K = \mathbb{Q}(\sqrt{\Delta}) = \mathbb{Q}(r_1)$ of $\mathbb{Q}$ of degree 2. The equal footing of the two roots gives rise to an F -fixing automorphism of K : $\sigma(p+qr_1) = p+qr_2$ that *permutes* r_1 and r_2 . We understand it as a *symmetry* of the root field K .

(Though inspired by the quadratic equations, we point out that the concepts of field isomorphism and automorphism are generic to any field, and the F -fixing automorphism is generic to any extension field K of F .)

But why do we care about such weirdly defined symmetry of the root field K in the first place? Does it have anything to do with the solvability of polynomials? If you have these questions in mind, you are hitting on the core of Galois theory. Let's try to get a taste of it through the following example.

4.7 A taste of Galois theory: F -fixing automorphisms and root permutations

Think of this artificial polynomial:

$$x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3) \text{ over } F = \mathbb{Q}.$$

Evidently, its four roots are $\pm\sqrt{2}$ and $\pm\sqrt{3}$, none in the coefficient field $F = \mathbb{Q}$, and its root field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ we discussed before.

¹⁴For example, rotating a square by 90° or reflecting it across a diagonal leaves the square geometrically unchanged.

¹⁵Please check appendix A.2 if you are unfamiliar with the meaning of a group.

Inspired by quadratic equations, you may expect that every permutation among the four roots corresponds to a $\mathbb{Q}$ -fixing automorphism, but that is not the case! Suppose there is such a $\mathbb{Q}$ -fixing automorphism σ sending $\sqrt{2}$ to $\sqrt{3}$, then the respecting of multiplication indicates $\sqrt{3} \cdot \sqrt{3} = \sigma(\sqrt{2}) \cdot \sigma(\sqrt{2}) = \sigma(\sqrt{2} \cdot \sqrt{2}) = \sigma(2)$, however, being $\mathbb{Q}$ -fixing means $\sigma(2) = 2$, giving the absurd result $3 = 2$.

Similar arguments disprove any map sending $\pm\sqrt{2}$ to $\pm\sqrt{3}$ or vice versa from being a $\mathbb{Q}$ -fixing automorphism.

On the other hand, by recalling every element in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ can be uniquely expressed as $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$, we can check [exercise] that the following two maps:

$$\alpha(a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}) = a - b\sqrt{2} + c\sqrt{3} - d\sqrt{6},$$

$$\beta(a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}) = a + b\sqrt{2} - c\sqrt{3} - d\sqrt{6},$$

are $\mathbb{Q}$ -fixing automorphisms¹⁶. In particular, the map α (map β) permutes roots $\pm\sqrt{2}$ (roots $\pm\sqrt{3}$) while leaving the other pair of roots unchanged.

Moreover, the composite map $\gamma = \alpha \circ \beta = \beta \circ \alpha$:

$$\gamma(a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}) = a - b\sqrt{2} - c\sqrt{3} + d\sqrt{6},$$

is also a $\mathbb{Q}$ -fixing automorphism.

4.8 A taste of Galois theory: implications of F -fixing automorphisms

In sum, among the total of $4! = 24$ possible permutations among the four roots, any of them involving permuting $\pm\sqrt{2}$ with $\pm\sqrt{3}$ cannot correspond to a $\mathbb{Q}$ -fixing automorphism of the root field, while the rest 4 of them: α , β , $\gamma = \alpha \circ \beta$, and the identity map ε are $\mathbb{Q}$ -fixing automorphisms.

As we mentioned, $\mathbb{Q}$ -fixing automorphisms of the extension field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are *symmetries* of it: the elements of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ *outside of* $\mathbb{Q}$ can roam under these automorphisms without causing “algebraic troubles” (breaking relations given by $+$, $-$, $\times$, $\div$). The automorphism α implies that two roots $\pm\sqrt{2}$ are on an equal footing (similarly, β implies $\pm\sqrt{3}$ are on an equal footing), while $\sqrt{2}$ and $\sqrt{3}$ are not on an equal footing.

Now we try solving the equation by radicals: constructing radical extensions of the coefficient field step-by-step to include all the roots.

First, we extend $\mathbb{Q}$ to $\mathbb{Q}(\sqrt{2})$, which includes two roots $\pm\sqrt{2}$ to our realm. Something interesting now happens: if we consider our coefficient field $F = \mathbb{Q}(\sqrt{2})$ ¹⁷, α , hence together

¹⁶If you wonder why we flip the sign in front of d , recall that in terms of roots $\sqrt{6} = \sqrt{2} \cdot \sqrt{3}$. You will get a better understanding in the next section.

¹⁷Of course we are allowed to do so, as $\mathbb{Q}(\sqrt{2})$ is a field containing all the coefficients!

with $\gamma = \alpha \circ \beta$, are no longer F -fixing automorphisms of the root field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. We are left with ε and β , and β implies the two roots $\pm\sqrt{3}$ can be swapped without touching $\mathbb{Q}(\sqrt{2})$ – $\pm\sqrt{3}$ are still on an equal footing after we extend to $\mathbb{Q}(\sqrt{2})$.

Next, we extend further to $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. The only F -fixing automorphism left is the identity map ε , indicating that now no roots can be swapped without touching F , that is, all the roots are in F – none of them are still on an equal footing: *all the roots are completely pinned down!*

From the above example, we observe that as we extend the coefficient field F : $\mathbb{Q} \rightarrow \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{2}, \sqrt{3})$ the corresponding F -fixing automorphisms of the root field¹⁸ become less and less: $\{\varepsilon, \alpha, \beta, \gamma\} \rightarrow \{\varepsilon, \beta\} \rightarrow \{\varepsilon\}$. Once F is extended enough to include all the roots, that is, F contains the root field, the only F -fixing automorphism of the root field we are left with is the identity map ε .

This observation sheds some light (to be constructed with rigor) on a strategy for extending F to include all the roots: we don't blindly extend F , but try to find (radical) extensions that can kick out some F -fixing automorphisms of the root field, one step at a time, until we are left only with ε . If this can be done, we know that our polynomial is *solvable by radicals*.

4.9 A taste of Galois theory: discussing root permutations without knowing roots?

But what we just mentioned is absurd: we don't know yet what the root field is before solving for all the roots! Hence we don't know what those automorphisms (root permutations) are, not to speak of kicking them out.

That's true, but remember our aim: we are not solving the polynomial, but only to know whether it is solvable. What if there is something we can know about the F -fixing automorphisms of the root field without knowing the roots, which can let us know whether the shrinking is possible?

So here comes the profound discovery of Galois: Recall that at each step of extension, the F -fixing automorphisms $(\{\varepsilon, \alpha, \beta, \gamma\} \rightarrow \{\varepsilon, \beta\} \rightarrow \{\varepsilon\})$ as in our example) form a *group*. Galois realized that these groups have internal connections that allow you to see, given the largest group at the beginning¹⁹, whether it *can be* shrunk down to $\{\varepsilon\}$ by radical extensions.

¹⁸You may want to emphasize that they are a special kind of F -fixing automorphisms *associated with root permutations*, but we will show in a second that they are actually not special – all F -fixing automorphisms are of this kind.

¹⁹How can we possibly know this group without knowing the roots? This is a great question. Keep it in mind! We will comment on it along the way.

4.10 A taste of Galois theory: summary of the recipe

In sum, given a polynomial $a(x)$ over its coefficient field F , the F -fixing automorphisms of its root field K , associated with certain root permutations, form a group. Galois pointed out properties of this group that can tell you whether it can be shrunk down all the way to the trivial group $\{\varepsilon\}$ by radical extensions of F . If it can be done, then the fact that the only F -fixing automorphism of K is the identity map indicates that all the roots are now in F , i.e., all the roots can be expressed by radicals in terms of the elements in the original coefficient field. Otherwise, $a(x)$ is not solvable by radicals.

You may have one last question here: can we know the group of F -fixing automorphisms at the beginning step, especially without knowing the root field? Generally no, but yes for the ones important to us, especially for the ultimate purpose of this drama – proving the non-existence of quintic formulas.

4.11 Intermission ending: Act II will begin...

Now that we have a vague impression what the palace looks like, the rest of the drama is to show why it stands solidly on the ground by revealing all the construction details – proving that the patterns we observed in the special examples are universally true.

As a sidenote: some of you may think that by definition, there can be fields that look nothing like $\mathbb{Q}$ or $\mathbb{R}$. For instance, $\{0, 1\}$ is a field with the zero element 0 and unit element 1, once we require $1 + 1 = 0$. In a generic field, adding its unit element repeatedly may result in 0. This is captured by the **characteristic** of a field: the least positive integer n such that the sum of n copies of the unit element is 0. If there is no such n , the field has **characteristic 0**, like our familiar $\mathbb{Q}, \mathbb{R}, \mathbb{C}$.

Galois theory is actually capable of dealing with any generic fields, but for fields with nonzero characteristics, a couple more points need to be addressed while not serving our purpose directly. We will from now on **confine our attention to fields of characteristic 0**, like $\mathbb{Q}, \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

5 Galois theory: the heart of the matter for solvability

5.1 F -fixing automorphisms and root permutations

We start by showing the deep connection between F -fixing automorphisms of a root field and the root permutations:

Let K be a root field of a polynomial $a(x) = a_0 + a_1x + \dots + a_nx^n$ over F . Denote $r_1, \dots, r_n$ as the roots of $a(x)$, then $K = F(r_1, \dots, r_n)$ ²⁰. For $a(r_k) = 0$, by taking a F -fixing automorphism σ on both sides, we get

$$\sigma(a(r_k)) = a_0 + a_1\sigma(r_k) + \dots + a_n[\sigma(r_k)]^n = \sigma(0) = 0.$$

Then we see that $\sigma(r_k)$ is also a root of $a(x)$, i.e., a F -fixing automorphism of K should always send a root of $a(x)$ to another root.

On the other hand, by (3.5), every $x \in K$ is a linear combination of $r_1^{i_1} \dots r_n^{i_n}$ with coefficients in F . Then $\sigma(x)$ for any x is completely determined by $\sigma(r_k)$ for every r_k , that is, the map σ is determined by how it permutes the n roots.

Now we see how F -fixing automorphisms of a root field are deeply connected with root permutations:

Theorem 5.1. *For the root field K of a polynomial $a(x)$ over F , every F -fixing automorphism of K is completely determined by a permutation of roots of $a(x)$.*

This is *very* important! It is also worth emphasizing again that *NOT* every permutation of roots is associated with a F -fixing automorphism of K , even if all the roots are outside F , as we saw earlier with the example of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

The subset of root permutations that give F -fixing automorphisms are the symmetries of K that are closely related to the problem of solvability of $a(x)$. We will pay close attention to these automorphisms from now on.

5.2 The Galois group

Let K be an extension field of F , and σ, τ be two F -fixing automorphisms of K , it is evident that the composite map $\sigma \circ \tau$ is also an automorphism of K that fixes F . Since map composition is associative, all the F -fixing automorphisms of K form a group²¹ with map composition $\circ$ being the group operation, leading to the core object in Galois theory:

Definition 5.1 (Galois group). *If K is the root field of a polynomial $a(x) \in F[x]$, the group of all the F -fixing automorphisms is called the **Galois group** of $a(x)$ over F , also called the Galois group of $K : F$, denoted as $\text{Gal}(K : F)$.*

²⁰Though not matter here, you can imagine all roots are not in F , since if $r_1 \in F$, you can factorize $a(x) = (x - r_1)b(x)$, and instead say K is the root field of $b(x)$ over F .

²¹The audience should check the other two group axioms [exercise] using the definition of automorphism: bijective map $K \rightarrow K$ respecting field operations.

It is crucial to keep in mind that a Galois group is defined for a field extension $K : F$ (or a polynomial $a(x)$ with specified coefficient field F), not for K alone or $a(x)$ without specifying the coefficient field. Moreover, the extension field K should be a root field over F .

5.3 Intermediate field, fixer, and fixfield

In the remainder of this drama we let the meaning of F and K remain fixed: F is an arbitrary field and K is the root field of some polynomial $a(x) \in F[x]$. The intermission suggests that we will extend step-by-step from F to K , investigating the field extensions and the corresponding Galois groups, so for our convenience we shall define a couple of concepts and notations:

Definition 5.2 (Intermediate field). *As suggested by the name, a field I such that $F \subseteq I \subseteq K$ is called an **intermediate field** between F and K . Since K is the root field of $a(x)$ over F , it is also the root field of $a(x)$ over every intermediate field I .*

Definition 5.3 (Fixer). *With each intermediate field I , we denote $I^* = \text{Gal}(K : I)$ as the Galois group of $K : I$, called the **fixer** of I .*

Evidently, for every intermediate field I , its fixer I^* is a subgroup of $\text{Gal}(K : F)$.

Conversely to the notion of a fixer, with each subgroup H of $\text{Gal}(K : F)$ we can associate the subfield of K containing all elements that are not changed by any automorphism $\sigma \in H$:

Definition 5.4 (Fixfield). *For each subgroup H of $\text{Gal}(K : F)$, the subfield of K :*

$$\{x \in K : \sigma(x) = x \text{ for every } \sigma \in H\}, \quad (5.1)$$

*is called the fixed field of H . For brevity and euphony we call it **fixfield**.*

You can verify in a trice [exercise] that it is indeed a subfield of K . It obviously contains F and is therefore one of the intermediate fields.

To recapitulate: every subgroup H of $\text{Gal}(K : F)$ fixes an intermediate field I , called the **fixfield** of H . Every intermediate field I is fixed by a subgroup H (usually denoted as I^*) of $\text{Gal}(K : F)$, called the **fixer** of I .

5.4 Galois correspondence

The concepts of fixer and fixfield *suggests* that there is a *one-to-one correspondence* between the subgroups of $\text{Gal}(K : F)$ and the fields intermediate between F and K , as indicated by the example of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. You may wonder why we said “suggests” – well, the loopholes are a bit tricky to see: on one hand, we haven’t shown that two distinct intermediate fields I_1, I_2 cannot have identical Galois groups; on the other hand, we haven’t shown that H is the Galois group of its fixfield.

Nevertheless, this correspondence is true and is the very heart of Galois theory, tying together properties of field extensions and subgroups:

Theorem 5.2 (Galois correspondence). *There is a one-to-one correspondence between the subgroups of $\text{Gal}(K : F)$ and the intermediate fields between F and K (a root field of F). This correspondence, which matches every subgroup with its fixfield (or equivalently, matches every intermediate field with its fixer), is called a **Galois correspondence**.*

It is worth emphasizing that everything we discuss above, especially the Galois group and correspondence, relies on the assumption that K is a root field of some polynomial $a(x)$ over F .

We defer the proof of Galois correspondence to Sec. 6 in order not to distract the audience from understanding the main structure. However, it is worth observing that larger subfields correspond to smaller subgroups:

$$I_2 \subseteq I_1 \quad \text{iff} \quad I_1^* \subseteq I_2^*$$

from our classic example $\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}$

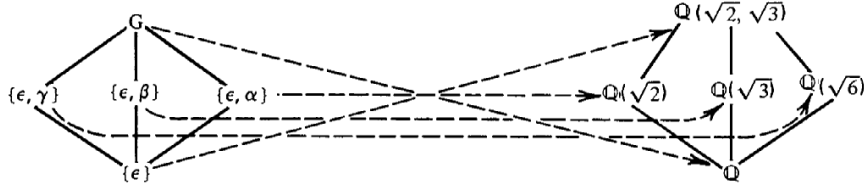


Figure 1. The Galois correspondence in $\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}$

Recall the criterion of solvability: there exists a radical extension E of the coefficient field F containing²² the root field K . The path from F to any radical extension E is a sequence of extensions $F \rightarrow I_1 \rightarrow I_2 \rightarrow \dots \rightarrow E$, where I_{i+1} is obtained by adjoining some root of an element in I_i . The crux of solvability is to understand the limitations of radical extensions – a restricted realm due to the connection between I_i and I_{i+1} .

As long as E is a root field of some polynomial over F , the Galois correspondence can translate the above sequence of field extensions to a chain of subgroups $\text{Gal}(E : F) \supseteq \text{Gal}(E : I_1) \supseteq \dots \supseteq \text{Gal}(E : E) = \{e\}$, reflecting the limitations of radical extensions in the language of group. As we will see, it turns out that checking whether $\text{Gal}(E : F)$ is compatible with the limitations in the language of group is much more practical than asking whether K can possibly be contained in some radical extension field E . This is Galois's approach to solvability.

²²Inspired by the $\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}$ example, you may think K itself is a radical extension of F when the polynomial is solvable, but that is generically not true.

5.5 Abelian extension and adjoining roots of unity

What is so special about the extension field $I_{i+1} = I_i(c)$, where c is a root of some element in I_i ? Let's first look at the n th roots of 1

Definition 5.5 (Roots of unity). *The **n th roots of unity** are the solutions of $x^n - 1 = 0$. By de Moivre's theorem in complex analysis^{P335–336}, they consist of a number ω and its first n powers:*

$$1 = \omega^0, \omega^1, \omega^2, \dots, \omega^{n-1},$$

where ω is called a **primitive n th root of unity**²³.

For any field F , $F(\omega) : F$ is an **Abelian extension** in the sense that $\text{Gal}(F(\omega) : F)$ is an **Abelian group**. Indeed, for g and h being two F -fixing automorphisms of $F(\omega)$, $g(\omega) = \omega^r$ and $h(\omega) = \omega^s$ for some integers r, s because an F -fixing automorphism should send roots to roots. Then $h \circ g(\omega) = \omega^{rs} = g \circ h(\omega)$. Using the definition of automorphisms and (3.3) you can easily show that $h \circ g = g \circ h$ [exercise].

5.6 Abelian extension and adjoining roots

Now suppose $\omega \in F$, so as all n th roots of unity. For $a \in F$ and a has an n th root b , then $F(b)$ is the root field of $x^n - a = 0$ over F , because all the n th root of a :

$$b, b\omega, b\omega^2, \dots, b\omega^{n-1}$$

are all in $F(b)$. In particular, $F(b) : F$ is an Abelian extension. Because F -fixing automorphisms of $F(b)$ sends roots to roots, for two F -fixing automorphisms $g(b) = b\omega^r$ and $h(b) = b\omega^s$

$$g \circ h(b) = g(b\omega^s) = b\omega^r\omega^s = h \circ g(b).$$

(Remember $\omega \in F$ is fixed.) Again $h \circ g = g \circ h$ follows by (3.3) [exercise].

So it is apparent that even though $F(b) : F$ obtained by adjoining an n th root b of $a \in F$ is not Abelian generically, we can make a buffer using the n th root of unity ω : $F \rightarrow F(\omega) \rightarrow F(\omega, b)$, such that both $F(\omega) : F$ and $F(\omega, b) : F(\omega)$ are Abelian.

5.7 Radical extension by a sequence of Abelian extensions

From the above observations the following important results arise:

Theorem 5.3 (Radical extension by a sequence of Abelian extensions). *For any field F , every radical extension field $E = F(c_1, \dots, c_n)$ of it is contained in a radical extension E_1 obtained by a sequence of Abelian extensions of adjoining n th roots of elements in the intermediate fields (which are radical themselves):*

$$F \rightarrow I_1 \rightarrow I_2 \rightarrow \dots \rightarrow E_1. \quad (5.2)$$

²³We say “a” primitive n th root of unity instead of “the” primitive n th root of unity because it is not unique. For instance, $e^{i\pi/3}$ and $e^{2i\pi/3}$ both satisfies the definition of a primitive 3rd root of unity.

The construction of E_1 is as follows: before adjoining c_i - an n th root - we first adjoin the primitive n th root of unity ω (if not already in the field). Or we can first adjoin all the needed roots of unity, and then adjoin the c_i 's. The resulting E_1 is the smallest field containing F , all the c_i 's, and all the needed roots of unity.

This is the key property of radical extensions for tackling solvability. As a reminder, $\text{Gal}(I_{i+1} : I_i)$ associated with (5.2) is **Abelian** at every step i .

5.8 Radical extension vs. root field

You may now perceive that by Galois correspondence, (5.2) corresponds to a chain of fixers $I_i^* = \text{Gal}(E_1 : I_i)$ - NOT YET! We need E_1 to be a root field of some polynomial over F for the Galois correspondence to kick in²⁴!

It is common that a radical extension is not a root field of any polynomial over the coefficient field.

Example 5.1. $\mathbb{Q}(\sqrt[3]{2})$ is not a root field of any polynomial over $\mathbb{Q}$: the other two complex roots of $x^3 - 2$ are missing. (This is not a trivial claim!)

Fortunately, we can always find a root field E_2 over F containing the radical extension E_1 in (5.2), which can also be obtained by a sequence of Abelian extensions of adjoining n th roots, for Galois correspondence to take effect. We defer the proof to Sec. 6.

5.9 Pause: what we have so far...

Let's take a break to see what we have done in the previous four subsections:

A polynomial is solvable by radicals if and only if there exists a radical extension E of its coefficient field F containing its root field K . For Galois correspondence to help us translate this condition into the language of groups, we showed (or claimed):

- Every radical extension E of F is contained in a radical extension E_1 obtained by a sequence of Abelian extensions.
- Every such E_1 is contained in a root field E_2 of some polynomial over F , which is radical over F and can be obtained by a sequence of Abelian extensions.

As a consequence, a polynomial $a(x) \in F[x]$ is solvable by radicals *if and only if* there exists a radical extension $E_2 : F$, such that E_2 contains the root field K of $a(x)$ and is a root field over F that can be obtained from F by a sequence of Abelian extensions.

With the crux of solvability reformulated into the existence of such an E_2 field, it is now perfectly suitable to apply Galois correspondence to translate the existence of such an

²⁴Refer to Sec. 5.4. You may think about the fixers $\text{Gal}(K : I_i)$ instead, but there is no way to guarantee that $I_i \subseteq K$ as we extend.

E_2 into the language of groups, which is much more practical to check.

You see how hard we've tried to make E_2 a root field and constructable by a sequence of Abelian extensions, not merely a radical extension of F – the former is essential for Galois correspondence to apply, while the latter indicates that for the sequence of Abelian extensions from F to E_2

$$F \rightarrow I_1 \rightarrow I_2 \rightarrow \dots \rightarrow E_2,$$

every fixer $I_i^* = \text{Gal}(E_2 : I_i)$ in the corresponding chain of subgroups

$$\text{Gal}(E_2 : F) \supseteq I_1^* \supseteq I_2^* \supseteq \dots \supseteq \text{Gal}(E_2 : E_2) = \{e\}$$

has a nice property due to the fact that $\text{Gal}(I_{i+1} : I_i)$ is Abelian.

5.10 Fundamental theorem of Galois theory

We hope to connect the Abelian nature of $\text{Gal}(I_{i+1} : I_i)$ with the fixers in

$$\text{Gal}(E_2 : F) \supseteq I_1^* \supseteq I_2^* \supseteq \dots \supseteq \text{Gal}(E_2 : E_2) = \{e\}.$$

Based on some theorems in group theory, Galois showed a key property of Galois groups:

Theorem 5.4. *Suppose $F \subseteq I \subseteq K$, where I and K are both root fields over F , then*

$$\text{Gal}(I : F) \cong \frac{\text{Gal}(K : F)}{\text{Gal}(K : I)}. \quad (5.3)$$

*It follows that $\text{Gal}(K : I)$ is a **normal subgroup**²⁵ of $\text{Gal}(K : F)$.*

This property, together with the Galois correspondence, is called **the fundamental theorem of Galois theory**. The proof is deferred to Sec. 6.

5.11 Solvability in terms of Galois groups

Applying the fundamental theorem to the sequence of extensions from F to E_2 :

$$F \rightarrow I_1 \rightarrow I_2 \rightarrow \dots \rightarrow E_2,$$

we get that for the corresponding chain of subgroups

$$\text{Gal}(E_2 : F) \supseteq I_1^* \supseteq I_2^* \supseteq \dots \supseteq \text{Gal}(E_2 : E_2) = \{e\},$$

each $I_{i+1}^* = \text{Gal}(E_2 : I_{i+1})$ is a *normal subgroup* of $I_i^* = \text{Gal}(E_2 : I_i)$, and the quotient group $I_i^*/I_{i+1}^* \cong \text{Gal}(I_{i+1} : I_i)$ is *Abelian*, according to (5.3).

In sum, if a polynomial $a(x) \in F[x]$ is solvable by radicals, then there exists a radical extension E_2 , being a root field of some polynomial over F containing the root field K of $a(x)$, such that the Galois group $\text{Gal}(E_2 : F)$ has a chain of subgroups

$$\text{Gal}(E_2 : F) \supseteq I_1^* \supseteq I_2^* \supseteq \dots \supseteq \text{Gal}(E_2 : E_2) = \{e\}$$

where each I_{i+1}^* is a normal subgroup of I_i^* (denoted as $I_{i+1}^* \triangleleft I_i^*$), and the quotient group I_i^*/I_{i+1}^* is Abelian.

²⁵You may feel punched by the weird symbols and terminologies above if you are not familiar with group theory. But don't get depressed! They are all explained in the appendix A.2 - A.14 and not hard to capture.

5.12 Solvable group

Solvability of $a(x) \in F[x]$ by definition is the existence of a radical extension E of F containing the root field K , and we just show it being equivalent to the existence of a radical extension E_2 of F , containing K while additionally being a root field over F and constructable by a sequence of Abelian extensions.

The additional properties let the fundamental theorem of Galois theory indicate that the Galois group $\text{Gal}(E_2 : F)$ is a **solvable group**:

Definition 5.6 (Solvable group). *A group G is called **solvable** if it has a chain of normal subgroups*

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_m = \{e\}$$

such that every quotient G_i/G_{i+1} is Abelian.

Therefore, $a(x)$ being solvable is *equivalent* to the existence of a radical extension E_2 of F containing the root field K of $a(x)$, such that the Galois group $\text{Gal}(E_2 : F)$ is a solvable group – a group-theoretic signature.

5.13 Galois’s criterion of solvability

Now you may want to protest: I don’t see how translating the crux of solvability into the above language of groups mitigates the difficulty at all! Indeed, originally we ask for a radical extension E containing K , now we ask for a radical extension E_2 containing K and in addition, $\text{Gal}(E_2 : F)$ being solvable, which seems even more complicated.

Well, the significance of the above criterion is not in the practical aspect, but is to serve as a stepping stone for proving:

Theorem 5.5 (Galois’s criterion of solvability). *A polynomial $a(x) \in F[x]$ is solvable by radicals if and only if its Galois group $\text{Gal}(K : F)$ is a solvable group (K is the root field of $a(x)$).*

To prove it, we first need to state and prove a lemma about solvable groups. The proof relies on a theorem (Theorem A.11) about quotient groups discussed in the appendix A.11: G/H is Abelian if and only if H contains all the **commutators** of G (elements of the form $xyx^{-1}y^{-1}$ for $x, y \in G$).

5.14 A lemma towards Galois’s criterion of solvability

Lemma 5.6. *Any **homomorphic image**²⁶ of a solvable group is solvable.*

Proof. Let G be a solvable group, with a chain of normal subgroups

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_m = \{e\}$$

²⁶Check appendix for group homomorphism and related stuff if you don’t know them.

such that every quotient G_i/G_{i+1} is Abelian. Let $f : G \rightarrow X$ be a homomorphism from G onto X , then clearly $X \supseteq f(G_1) \supseteq \cdots \supseteq \{e\}$, and every $f(G_i)$ is a normal subgroup of $f(G_{i-1})$ because if $f(a) \in f(G_i)$ and $f(x) \in f(G_{i-1})$, then $a \in G_i$ and $x \in G_{i-1}$; hence, $xax^{-1} \in G_i$ (since G_i is normal in G_{i-1}) and $f(x)f(a)f(x)^{-1} = f(xax^{-1}) \in f(G_i)$.

Moreover, every quotient $f(G_{i-1})/f(G_i)$ is Abelian because for any $x, y \in G_{i-1}$, $xyx^{-1}y^{-1} \in G_i$ (since G_{i-1}/G_i is Abelian) and therefore $f(x)f(y)f(x)^{-1}f(y)^{-1} = f(xyx^{-1}y^{-1}) \in f(G_i)$. In sum, X is solvable. $\square$

5.15 Proof: Galois's criterion of solvability

We are now ready to prove Galois's criterion of solvability. To your disappointment, we will prove only the “only if” part: if $a(x)$ is solvable, then its Galois group $\text{Gal}(K : F)$ is solvable, because its contraposition – if $\text{Gal}(K : F)$ is not solvable, then $a(x)$ is not solvable – is what we need to disprove the solvability of generic quintic polynomials.

Proof. $a(x)$ is solvable means a radical extension $E_2 \supseteq K$ such that $\text{Gal}(E_2 : F)$ is solvable. By (5.3), $\text{Gal}(K : F) \cong \text{Gal}(E_2 : F)/\text{Gal}(E_2 : K)$, that is, $\text{Gal}(K : F)$ is a homomorphic image of the solvable group $\text{Gal}(E_2 : F)$, which is therefore solvable by Lemma 5.6. $\square$

To prove the if part, one needs to show that for $\text{Gal}(K : F)$ being solvable, K may be further extended^{P339,343–344} to a radical extension of F .

5.16 No quintic formula: S_5 as the Galois group

With Galois's criterion in hand, we are finally ready to disprove the solvability of generic quintic polynomials. It is not hard to perceive that the Galois group of a *generic* quintic polynomial

$$a(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + x^5$$

is S_5 : the group of all permutations of the five roots, because every permutation should be associated with a F -fixing automorphism of the root field unless there is some special relation between the roots²⁷ as a result of some coincidence in the coefficients, which is not the case for a *generic* polynomial.

If you are not comfortable with this, it is not hard to show that the Galois group of some particular quintic polynomials (e.g., $x^5 - 5x + 2$ ^{P339–341}) is exactly S_5 .

5.17 No quintic formula: S_5 is not solvable

It is very interesting to prove that S_5 is not solvable by contradiction.

Theorem 5.7. *The group S_5 is not solvable.*

²⁷As a trivial example: the Galois group of $(x-1)(x-2)^2(x-3)^2$ over $\mathbb{Q}$ is $\{e\}$, since all its roots are already in $\mathbb{Q}$.

Proof. Suppose S_5 is solvable, $S_5 \triangleright G_1 \triangleright G_2 \triangleright \cdots \triangleright G_m = \{e\}$ with every quotient G_i/G_{i+1} being Abelian. Then G_1 contains all the commutators of S_5 . It is evident that S_5 contains all the cycles of length 3, that is, the cyclic permutations $(i, j, k) \rightarrow (j, k, i)$ where i, j, k are three of the five roots, denoted as (ijk) . Recall from Sec. 5.13 that for every pair $x, y \in S_5$, their commutator $xyx^{-1}y^{-1} \in G_1$ is needed for S_5/G_1 to be Abelian. However, with the presence of 5 roots, every cycle (ijk) can be represented as a commutator²⁸

$$(ijk) = (kmj)(lji)(jmk)(ijl) = (kmj)(lji)(kmj)^{-1}(lji)^{-1}, \quad (5.4)$$

Therefore, G_1 contains all cycles of length 3. Following this line of reasoning, $G_2, \dots, G_m$ should also contain all cycles of length 3, contradicting to $G_m = \{e\}$. $\square$

5.18 No quintic formula: conclusion

Provided that S_5 , the Galois group of at least some quintic polynomials, is not solvable, those quintic polynomials are not solvable by radicals according to Galois's criterion of solvability.

This is to say that given a (characteristic zero) field F like $\mathbb{Q}, \mathbb{R}$, for some quintic polynomial $a(x)$ over F , not all of its five roots can be expressed in terms of radicals of elements in F .

Since all the coefficients of $a(x)$ are in F , the above indicates that some of its roots cannot be expressed in terms of radicals involving its coefficients and other elements in F .

In conclusion, there is no general radical formula that can express all the roots of every quintic polynomial taking the coefficients as variables:

Theorem 5.8 (Abel-Ruffini theorem). *There is no solution in radicals to general polynomial equations of degree five or higher with arbitrary coefficients.*

²⁸Videos demonstrating this fact: [Youtube \(English\)](#), [Bilibili \(Chinese\)](#).

6 The proofs!

6.1 The proofs we skipped

We have shown you the core structure of Galois theory, outlining its approach to the solvability problem of polynomials. However, as we indicated, to keep the audience focus on the mainstream we have skipped the proofs of some important results, including:

1. The fundamental theorem of Galois theory:
 - The Galois correspondence: the one-to-one correspondence between subgroups of $\text{Gal}(K : F)$ and intermediate fields between F and K , provided that K is a root field of some polynomial over F .
 - The key property of Galois groups: $\text{Gal}(I : F) \cong \text{Gal}(K : F)/\text{Gal}(K : I)$ for $F \subseteq I \subseteq K$, where I and K are both root fields over F .
2. For every radical extension E_1 of F constructed by a sequence of Abelian extensions as in (5.2), the existence of a root field $E_2 \supseteq E_1$, which is a radical extension of F itself and constructable by a sequence of Abelian extensions. (Let's refer to it as the E_2 -theorem for brevity.)

To prove them, we will need go through a dozen of theorems in the last three chapters of Pinter's book.

6.2 Disclaimer about the theorems and proofs

Before diving into the theorems and proofs, we would like to emphasize that they are included only for the sake of completeness. They will seem rather technical and not very enlightening for general audience. We encourage you to skip them (go directly to Sec. 7) if you are just here to see *how* Galois theory works, and not to care too much about *why* it works.

6.3 Irreducible polynomials have no multiple roots

We confined our attentions to characteristic 0 fields at the end of the intermission, because they have the following nice property:

Theorem 6.1. *If F is a field of characteristic zero, irreducible polynomials over F can never have multiple roots.*

(A polynomial $a(x) \in F[x]$ is **irreducible** if it cannot be factored into $a(x) = b(x)c(x)$ for some non-constant $b(x), c(x) \in F[x]$.)

Proof. Let's begin with a lemma: an irreducible polynomial $a(x)$ is a minimal polynomial of any of its roots.

Suppose $a(x) \in F[x]$ is an irreducible polynomial with a root c . Let $p(x)$ be the (monic) minimal polynomial of c over F . Due to the *division algorithm*^{P245–246}, $a(x) = m(x)p(x) + r(x)$ with $m(x), p(x), r(x) \in F[x]$ and $r(x)$ having lower degree than $p(x)$. Since

$p(c) = 0$, then $a(c) = 0$ implies $r(c) = 0$, so $a(x) = m(x)p(x)$. However, $a(x)$ is irreducible, so $m(x)$ must be a constant, and thus $a(x)$ is a minimal polynomial of c over F . This proves the lemma.

Now, suppose c is a multiple root of $a(x) = a_0 + a_1x + \dots + a_nx^n$, we have $a(x) = (x - c)^2b(x)$ for some $b(x) \in F[x]$. By the product rule for derivatives, $a'(x) = 2(x - c)b(x) + (x - c)^2b'(x)$, which also has c as a root – $a'(x)$ is also a multiple of $p(x)$ by a similar argument as above (the multiplier need not be a constant this time).

Recall that we have shown that $a(x)$ is a minimal polynomial of c , then $a'(x) = q(x)a(x)$ for some $q(x) \in F[x]$. But this can only happen if $a'(x) = q(x) = 0$, provided that $a'(x)$ has lower degree than $a(x)$. This implies that the leading coefficient na_n of $a'(x) = a_1 + 2a_2x + \dots + na_nx^{n-1}$ must be zero.

Here is where the characteristic zero condition comes into play: for $a_n \neq 0$ and $n \neq 0$ in F , then $na_n \neq 0$ for our characteristic zero F – we get a contradiction. Therefore, $a(x)$ cannot have multiple roots. $\square$

6.4 Every finite extension of a field is a simple extension

At the end of Sec. 3, we mentioned that every finite extension is an iterated extension (and vice versa). Now, with characteristic 0 field F , even more is true:

Theorem 6.2. *Every finite extension of F is a simple extension.*

Proof. We only need to show that any iterated extension like $F(a, b)$ is equal to $F(c)$ for some c .

Let $A(x)$ and $B(x)$ be the minimal polynomials of a and b over F , respectively. Since $A(x)$ and $B(x)$ are irreducible, they have distinct roots $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_m$. Let t be any element of F such that

$$t \neq \frac{a_i - a_1}{b_1 - b_j}, \quad \forall i, j \neq 1.$$

Denoting $c = a_1 + tb_1$, it follows that $c \neq a_i + tb_j$, or equivalently $c - tb_j \neq a_i$ for any $i, j \neq 1$. We will show that $F(a_1, b_1) = F(c)$.

Define $h(x) = A(c - tx)$, then $h(x)$ has b_1 as a root because $h(b_1) = A(c - tb_1) = A(a_1) = 0$. Moreover, $h(b_j) = A(c - tb_j) \neq 0$, because $c - tb_j \neq a_i$ for any i – b is the only common root of $h(x)$ and $B(x)$.

We will prove $b_1 \in F(c)$, and hence $a_1 = c - tb_1 \in F(c)$ (remember that $t \in F$), giving rise to $F(a_1, b_1) \subseteq F(c)$. Since $c = a_1 + tb_1 \in F(a_1, b_1)$, we have $F(c) \subseteq F(a_1, b_1)$, and thus $F(a_1, b_1) = F(c)$.

It remains only to prove that $b_1 \in F(c)$. Let $p(x)$ be the minimal polynomial of b_1 over $F(c)$. If $p(x)$ has degree 1, then $p(x) = x - b_1$, so $b_1 \in F(c)$ and we are done. Now suppose $p(x)$ has degree greater than 1, then both $h(x)$ and $B(x)$ are multiples of $p(x)$, because they both have b_1 as a root (refer to the proof of the previous theorem). However, this contradicts to the fact that b_1 is the only common root of $h(x)$ and $B(x)$, since they must share all (no less than 2) the roots of $p(x)$. Therefore, $p(x)$ has degree 1 and $b_1 \in F(c)$. $\square$

6.5 Further connections between F -fixing automorphisms and root permutations

We showed a deep relation at the very beginning of Sec. 5: every F -fixing automorphism of the root field K of a polynomial $a(x) \in F[x]$ corresponds to a permutation of its roots. The fact that irreducible polynomials (over characteristic 0 fields) have no multiple roots will allow us to say more about this connection and get to know more about the features of root fields, which will be useful for the proofs of the fundamental theorem of Galois theory and Galois's criterion of solvability.

The trail towards this goal is, in fact, a bit rugged and protracted, filled with seemingly pointless concepts and theorems, though they will all add up at the end of the day. So please refresh your mind and keep up.

(The fact that every finite extension is simple will play a role in the following discussion.)

Definition 6.1 (Extension of isomorphism). *Suppose F_1 and F_2 are two fields, and $h : F_1 \rightarrow F_2$ is an isomorphism between them. Let K_1 and K_2 be extension fields of F_1 and F_2 respectively, and $\bar{h} : K_1 \rightarrow K_2$ also be an isomorphism. We call $\bar{h}$ an **extension**²⁹ of h if $\bar{h}(a) = h(a)$ for every $a \in F_1$.*

An important example of isomorphism extension arises from the fact that $F[x]$ is an extension³⁰ of F when we think of the elements in F as constant polynomials. Then given an isomorphism $h : F_1 \rightarrow F_2$, we can extend it to an isomorphism $\bar{h} : F_1[x] \rightarrow F_2[x]$ by defining

$$\bar{h}(a_0 + a_1x + \dots + a_nx^n) = h(a_0) + h(a_1)x + \dots + h(a_n)x^n$$

Verifying $\bar{h}$ is indeed an isomorphism (respecting addition and multiplication) is child's play [exercise].

For the sake of simplicity we will denote the polynomial $\bar{h}(a(x))$ as $ha(x)$, for one to imagine h maps every the coefficient a_i of $a(x)$ to $h(a_i)$. Because $\bar{h}$ is an isomorphism, $a(x)$ is irreducible over F_1 if and only if $ha(x)$ is irreducible over F_2 .

²⁹In the plain sense that it is formed by "adding on" to h .

³⁰Sometimes $F[x]$ is not a field but an integral domain, but it does not matter here.

Theorem 6.3. Suppose a and b are two roots of an irreducible polynomial $p(x) \in F[x]$, then there is an isomorphism $g : F(a) \rightarrow F(b)$ such that $g(a) = b$ and $g(y) = y$ for every $y \in F$.

Proof. Every element in $F(a)$ is of the form $c_0 + c_1a + \dots + c_na^n$ where $c_i \in F$, and every element in $F(b)$ is of the form $d_0 + d_1b + \dots + d_nb^n$ where $d_i \in F$ ³¹. We can define $g : F(a) \rightarrow F(b)$ by

$$g(c_0 + c_1a + \dots + c_na^n) = c_0 + c_1b + \dots + c_nb^n.$$

It is clear that $g(a) = b$. Verifying g is an isomorphism (i.e., it preserves addition and multiplication) is straightforward [exercise^{P315,321}]. $\square$

As you may already notice, the above theorem is a special case of the following more general result:

Theorem 6.4. Let $h : F_1 \rightarrow F_2$ be an isomorphism, and let $p(x)$ be irreducible over F_1 (hence $hp(x)$ is irreducible over F_2). Suppose a is a root of $p(x)$ and b is a root of $hp(x)$, then h can be extended to an isomorphism $\bar{h} : F_1(a) \rightarrow F_2(b)$ such that $\bar{h}(a) = b$.

Proof. The proof is similar to the previous one, except that we need to define $\bar{h}$ on $F_1(a)$ by

$$\bar{h}(c_0 + c_1a + \dots + c_na^n) = h(c_0) + h(c_1)b + \dots + h(c_n)b^n.$$

Verifying $\bar{h}$ is an isomorphism is also straightforward [exercise^{P315,321}]. Please note that for an automorphism h , $h(1) = 1$, then $\bar{h}(a) = \bar{h}(1a) = h(1)b = b$. $\square$

With the above facts about generic isomorphism extension, we can now proceed to discuss isomorphisms/automorphisms of root fields.

Let's first consider the following situation: K and K' are finite extensions of F , having a common extension E . If $h : K \rightarrow K'$ is an isomorphism such that $h(y) = y$ for every $y \in F$, we say h **fixes** F . Let $c \in K$ be a root of some polynomial $a(x) = a_0 + \dots + a_nx^n \in F[x]$, then $h(c)$ is also a root of $a(x)$, because the coefficients of $a(x)$ are fixed by h :

$$a(h(c)) = a_0 + a_1h(c) + \dots + a_nh(c)^n = h(a(c)) = h(0) = 0.$$

This is an important fact: for any $a(x) \in F[x]$ ($a(x)$ needs not being irreducible), every F -fixing isomorphism sends roots of $a(x)$ to roots of $a(x)$.

Now, if K is a root field of $a(x)$, say $K = F(c_1, \dots, c_n)$ where c_i are the roots of $a(x)$, then any F -fixing isomorphism $h : K \rightarrow K'$ must send each root c_i to another root of $a(x)$, that is, *permutes* $c_1, \dots, c_n$. However, we know that every element in K is a sum of the terms like $kc_1^{i_1}c_2^{i_2}\dots c_n^{i_n}$ where $k \in F$. Then h sends the product $c_1^{i_1}c_2^{i_2}\dots c_n^{i_n}$ to another product of the same form. That is to say, h sends every element in K to another element

³¹The degrees $[F(a) : F] = [F(b) : F]$, since the irreducible polynomial $p(x)$ is a minimal polynomial for each of its roots.

in K . In other words, K' is the same field as K , and h is an automorphism of K .

We summarize the above observation in the following theorem:

Theorem 6.5. *Let K and K' be finite extensions of F . Assume K is the root field of some polynomial over F . If $h : K \rightarrow K'$ is an isomorphism which fixes F , then $K' = K$ and h is an F -fixing automorphism of K .*

This theorem is often used in tandem with a stronger version of it:

Theorem 6.6. *Let L and L' be finite extensions of F . Let K be an extension of L such that K is a root field over F . Any isomorphism $h : L \rightarrow L'$ which fixes F can be extended to an F -fixing automorphism $\bar{h} : K \rightarrow K$.*

Proof. By Theorem 6.2, K is a simple extension of L , say $K = L(c)$. Then we can make use of Theorem 6.4 to extend h to an isomorphism $\bar{h} : L(c) \rightarrow L'(d) = K'$. Finally, by Theorem 6.5, $K' = K$. $\square$

It follows that $L' \subseteq K$ since the range of h is a subfield of the range of $\bar{h}$.

We can combine this above theorem with Theorem 6.3 (take $L = F(a)$ and $L' = F(b)$) to get:

Theorem 6.7. *If a and b are two roots of an irreducible polynomial $p(x) \in F[x]$, there is an F -fixing automorphism of its root field K sending a to b .*

We now present two strong properties of root fields:

Theorem 6.8. *Let K be the root field of some polynomial over F . For **every** irreducible $p(x) \in F[x]$ having one root in K , then all roots of $p(x)$ are in K .*

Proof. Let $a \in K$ be a root of $p(x)$, and b be another root of $p(x)$. By Theorem 6.3 there is an isomorphism $g : F(a) \rightarrow F(b)$ fixing F . But $F(a) \subseteq K$, so Theorem 6.6 (and the following conclusion $L' \subseteq K$) indicates $F(b) \subseteq K$, and thus $b \in K$. $\square$

Theorem 6.9. *Let K be the root field of some polynomial over F . The number of F -fixing automorphisms of K is equal to $[K : F]$.*

Proof. Suppose $K = F(a)$ (Theorem 6.2), $[K : F] = n$ is the degree of the minimal polynomial $p(x)$ of a over F . By Theorem 6.7 there is a distinct F -fixing automorphism of K sending a to each of the n distinct roots of $p(x)$ in K . Moreover, every F -fixing automorphism of $F(a)$ is pinned down by where it sends a . Therefore, there are exactly $n = [K : F]$ such automorphisms. $\square$

6.6 Proving Galois correspondence

We are finally ready to prove the Galois correspondence: The *one-to-one correspondence* between the subgroups of $\text{Gal}(K : F)$ and the intermediate fields between F and K , provided that K is a root field over F . This can be split into the next two theorems.

Theorem 6.10. *If H is the fixer of I ($H = \text{Gal}(K : I)$, the group of all I -fixing automorphisms of K), then I is the fixfield of H .*

Proof. It might appear first sight that I is the fixfield of H because every automorphism in H fixes elements in I . However, we need to show that there is no element outside I that is fixed by every automorphism in H (by the definition of fixfield).

Let's suppose I' is the fixfield of H and prove that $a \notin I$ implies $a \notin I'$. The minimal polynomial $p(x) \in I[x]$ of $a \notin I$ must have degree ≥ 2 , then $p(x)$ has another root $b \neq a$. By Theorem 6.7, there is an I -fixing automorphism of K in H sending a to b , then $a \notin I'$. $\square$

To complete the proof of Galois correspondence, we have to show the converse:

Theorem 6.11. *If I is the fixfield of H , then H is the fixer of I , i.e., $H = \text{Gal}(K : I)$.*

It is clear from the definitions of fixer and fixfield that every automorphism in H fixes I . We need to show that there is no automorphism outside H that fixes I .

We will prove it by showing that there are as many elements in H as in $\text{Gal}(K : I)$, which requires a lemma:

Lemma 6.12. *Let H be a subgroup of $\text{Gal}(K : F)$, and I be the fixfield of H . The number of elements in H is equal to $[K : I]$.*

Proof. Let H have r elements $h_1, h_2, \dots, h_r$, and $K = I(a)$ (by Theorem 6.2). Construct the polynomial

$$b(x) = (x - h_1(a))(x - h_2(a)) \dots (x - h_r(a)).$$

A surprising result is that actually $b(x) \in I[x]$. To see this, recall from the discussion under Definition 6.1 that every automorphism $h_i : K \rightarrow K$ can be extended to $\bar{h}_i : K[x] \rightarrow K[x]$, and we get

$$\begin{aligned} \bar{h}_i(b(x)) &= \bar{h}_i(x - h_1(a))\bar{h}_i(x - h_2(a)) \dots \bar{h}_i(x - h_r(a)) \\ &= (x - h_i \circ h_1(a))(x - h_i \circ h_2(a)) \dots (x - h_i \circ h_r(a)). \end{aligned}$$

But $h_i \circ h_1, h_i \circ h_2, \dots, h_i \circ h_r$ are just a rearrangement of $h_1, h_2, \dots, h_r$, because H is a group ($h_i \circ h_j = h_i \circ h_k$ iff $h_j = h_k$). Therefore, $\bar{h}_i(b(x)) = b(x)$ for every i , meaning that every coefficient of $b(x)$ is left unchanged by every $h_i \in H$ – the coefficients are in the fixfield I of H and thus $b(x) \in I[x]$.

Moreover, $b(a) = 0$ since one of the h_i is the identity map, so $b(x)$ is a nonzero polynomial in $I[x]$ having a as a root. By the proof of Theorem 6.1, $b(x)$ is a multiple of the minimal polynomial of a over I , which has degree $[I(a) : I] = [K : I]$. Since $b(x)$ has degree r , we have $r \geq [K : I]$. On the other hand, Theorem 6.9 says that there are $[K : I]$ I -fixing automorphisms of K in total, so $r \leq [K : I]$. Therefore, $r = [K : I]$. $\square$

With the above lemma, the proof of Theorem 6.11 is a breeze³².

Proof. With I being the fixfield of H , let's say H have r elements, then $r = [K : I]$ by Lemma 6.12. We know from Theorem 6.10 that I is the fixfield of $\text{Gal}(K : I)$, and Lemma 6.12 again tells us that $\text{Gal}(K : I)$ should have $[K : I]$ elements, so H and $\text{Gal}(K : I)$ have the same number of elements. Since $H \subseteq \text{Gal}(K : I)$ by definition, we must have $H = \text{Gal}(K : I)$. $\square$

After walking through such a tedious trail, we have finally proved the Galois correspondence, which is a half of the fundamental theorem of Galois theory. The other half, a key property of Galois group: $\text{Gal}(I : F) \cong \text{Gal}(K : F) / \text{Gal}(K : I)$ for $F \subseteq I \subseteq K$, where I and K are both root fields over F , will be proved next.

6.7 Proving $\text{Gal}(I : F) \cong \text{Gal}(K : F) / \text{Gal}(K : I)$

Proof. Suppose $F \subseteq I \subseteq K$, where I and K are both root fields over F . (It is evident that K is a root field over I as well.) For an F -fixing automorphism h of K , we can **restrict** it to a smaller domain I , called a **restriction** of h . It is clearly an isomorphism of I fixing F , and by Theorem 6.5, it is an F -fixing automorphism of I .

Now, with $h \in \text{Gal}(K : F)$ and its restriction to I in $\text{Gal}(I : F)$, we can define a map $\mu : \text{Gal}(K : F) \rightarrow \text{Gal}(I : F)$ by

$$\mu(h) = \text{the restriction of } h \text{ to } I.$$

It is easy to verify that μ is a **group homomorphism** (see appendix A.7 and after for details), and it is surjective because every F -fixing automorphism of I can be extended to an F -fixing automorphism of K by Theorem 6.6 (taking $L = L' = I$).

Finally, if $h \in \text{Gal}(K : F)$, $\mu(h)$ is the identity map of I if and only if $h(x) = x$ for every $x \in I$, that is, h fixes I . This means the **kernel**³³ of μ is exactly $\text{Gal}(K : I)$ – the set of all the I -fixing automorphisms of K .

Now, we have a *surjective* group homomorphism $\mu : \text{Gal}(K : F) \rightarrow \text{Gal}(I : F)$ with kernel $\text{Gal}(K : I)$ – a setup being perfectly suitable for applying the **fundamental homomorphism theorem** (FHT) of group theory (see appendix A.12 – A.13 for details):

Theorem 6.13 (Fundamental homomorphism theorem of group theory). *Let $f : G \rightarrow H$ be a homomorphism of group G **onto** (being surjective) group H . If K is the kernel of f , then G/K (the quotient group of G by K) is isomorphic to H : $H \cong G/K$.*

³²Actually, provided that Theorem 6.10 already claims that I is the fixfield of $\text{Gal}(K : I)$, to prove Theorem 6.11, we only need to show that I is the fixfield of *only one* subgroup H of $\text{Gal}(K : F)$, which is then inevitably $\text{Gal}(K : I)$. This is accomplished by the fact that $\text{Gal}(K : I)$ contains *all* the I -fixing automorphisms of K (then H must be a subgroup of $\text{Gal}(K : I)$) and Lemma 6.12 that pins down the number of elements in H .

³³For a group homomorphism $f : G \rightarrow H$, its kernel is the set of all elements in G that are sent by f to the neutral element of H .

Applying FHT to μ , we get $\text{Gal}(I : F) \cong \text{Gal}(K : F) / \text{Gal}(K : I)$, which, together with the proof of Galois correspondence, completes the proof of the fundamental theorem of Galois theory. $\square$

6.8 The E_2 -theorem

The last thing we need to prove is the E_2 -theorem, which claims the existence of a root field E_2 over F , containing the root field K of the solvable polynomial of interest, being a radical extension of F , and being constructable by a sequence of Abelian extensions.

Before diving into the proof, let's recall the significance of E_2 , since it is the most nonintuitive concept in the whole story. In the intermission, we established the intuition that for the sequence of radical extensions $\mathbb{Q} \rightarrow \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{2}, \sqrt{3})$, the corresponding Galois groups $\{\varepsilon, \alpha, \beta, \gamma\} \rightarrow \{\varepsilon, \beta\} \rightarrow \{\varepsilon\}$ shrink down to the trivial group.

This intuitive picture seems to align perfectly with the generic idea of Galois correspondence, until we realize that this illustrative example is too special: the root field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ of the polynomial itself is a radical extension of the coefficient field $\mathbb{Q}$.

For a generic polynomial, when its root field K is not a radical extension of F , we will have to radically extend F step-by-step (recall the requirements we discussed at (5.2)) to a field E_1 containing K , not to K itself. But at this point, E_1 is not guaranteed to be a root field over F , so that the concept of Galois group “ $\text{Gal}(E_1 : F)$ ” is not applicable. Then we need to further extend E_1 to a field E_2 that is a root field over F , and thus has a Galois group $\text{Gal}(E_2 : F)$, and we proved that $\text{Gal}(E_2 : F)$ being solvable is a sufficient and necessary condition for the polynomial of interest to be solvable by radicals.

Till now, there is still a last piece of puzzle: we don't know anything about E_2 in the first place, not to mention the Galois group $\text{Gal}(E_2 : F)$, so how can we check the solvability of $\text{Gal}(E_2 : F)$? The fundamental theorem of Galois theory kicks in here, showing the equivalence of $\text{Gal}(E_2 : F)$ being solvable and $\text{Gal}(K : F)$ being solvable.

Although we don't know much of K as well, it is at least the root field of the polynomial of interest, so that $\text{Gal}(K : F)$ consists of permutations of the roots. This makes $\text{Gal}(K : F)$ accessible in a few crucial cases (e.g., for $x^5 - 5x - 2$ on Pinter's book^{P340}), even without knowing K .

6.9 Proving the E_2 -theorem

Fair enough for the review, let's get to the proof of the E_2 -theorem. This last proof of the entire drama is gonna be the longest among all the proofs we have encountered earlier, but we promise that it is not hard to follow.

Proof. Referring back to the discussion around (5.2), E_1 contains all the needed³⁴ n th roots of unity. Without losing generality, we adjoin all the needed primitive roots of unity at the very beginning. That is, we first extend the coefficient field F of our interested polynomial to a field $\tilde{F}$ by adjoining every primitive root of unity needed later in the sequence from F to E_1 . Evidently, $\tilde{F}$ can be obtained from F by a sequence of Abelian extensions (adjoining one primitive root of unity at a time, thus $\tilde{F}$ is a radical extension of F).

Moreover, $\tilde{F}$ is a root field over F : for $\tilde{F} = F(\omega_n, \omega_m, \dots)$ where ω_n is a primitive n th root of unity, then $\tilde{F}$ is the root field of the polynomial $(x^n - 1)(x^m - 1) \dots$ over F .

With all needed roots of unity in $\tilde{F}$, the sequence towards any radical extension of $\tilde{F}$ is a sequence of Abelian extensions: at each step, we adjoin an n th root of an element in the previous field, which is Abelian according to Sec. 5.6. In sum, the sequence of Abelian extensions from F to E_1 can be rearranged to

$$F \rightarrow F(\omega_n) \rightarrow F(\omega_n, \omega_m) \rightarrow \dots \tilde{F} \rightarrow \tilde{F}(c_1) \rightarrow \tilde{F}(c_1, c_2) \rightarrow \dots \rightarrow E_1 \quad (6.1)$$

where each c_i is an root of some element in the previous field.

Our aim is to prove that the final extension field E_1 in (5.2) is contained in a root field E_2 over F which is itself a radical extension of F and can be obtained by a sequence of Abelian extensions, but it is instructive to notice that E_1 is not special at all compared to the intermediate fields the above sequence (6.1). Therefore, proving the E_2 -theorem transforms into proving that every field in (6.1) has this property: contained in a root field over F which is a radical extension of F and can be obtained by a sequence of Abelian extensions.

It is apparent that we should prove it by induction. We have already shown $\tilde{F}$ and fields to its left in (6.1) bear this property. Now, suppose $F_i \equiv \tilde{F}(c_1, \dots, c_i)$ has this property; we want to show that $F_{i+1} \equiv \tilde{F}(c_1, \dots, c_i, c_{i+1})$, where c_{i+1} is a q th root of $a \in F_i$, has this property as well³⁵.

Since F_i has the property, it is contained in the root field of a polynomial $B_i(x) \in F[x]$. Let $H = \{h_1, \dots, h_r\}$ be the group of all the F -fixing automorphisms of F_i , and consider the polynomial

$$b_{i+1}(x) = (x^q - h_1(a))(x^q - h_2(a)) \dots (x^q - h_r(a)).$$

As in the proof of Lemma 6.12, $b_{i+1}(x)$ is a polynomial in $F[x]$ (all its coefficients are in the fixfield F of H) and has c_{i+1} as a root.

This amounts to saying that F_{i+1} is contained in the root field of $B_{i+1}(x) = B_i(x)b_{i+1}(x)$ over F – the root field of $B_{i+1}(x)$ contains the root field of $B_i(x)$ and the wanted root c_{i+1}

³⁴The “need” is specified by every step in the sequence of Abelian and radical extensions $F \rightarrow I_1 \rightarrow \dots \rightarrow E_1$: for adjoining an n th root of an element in I_i to be an Abelian extension, we need the corresponding primitive n th root of unity in I_i .

³⁵The proof on Pinter’s book seems incorrect from here...

of $b_{i+1}(x)$.

Moreover, the root field of $B_{i+1}(x)$ is a radical extension of F , because all the roots of $b_{i+1}(x)$ are q th roots of elements $h_1(a), \dots, h_r(a)$ in F_i , so one can obtain the root field of $B_{i+1}(x)$ from the root field of $B_i(x)$ by adjoining these q th roots one by one. Together with the fact that the root field of $B_i(x)$ is a radical extension of F by the induction hypothesis, we conclude that the root field of $B_{i+1}(x)$ is a radical extension of F as well.

Finally, since the induction hypothesis says that there is a sequence of Abelian extensions from F to the root field of $B_i(x)$, by continue adjoining the q th roots of $h_1(a), \dots, h_r(a)$ to it one-by-one we extend this sequence of Abelian extensions to the root field of $B_{i+1}(x)$.

As a result, we manage to show that F_{i+1} is contained in the root field of $B_{i+1}(x)$ over F , which is a radical extension of F and can be obtained by a sequence of Abelian extensions. Therefore, every field in the sequence from F to E_1 , including E_1 itself, has this property, proving the E_2 -theorem. $\square$

7 End Matter

We believe the audience now see how Galois theory provides a *sufficient and necessary* condition for the solvability of polynomials in terms of the properties of their Galois groups, and how it is used to disprove the solvability of generic quintic polynomials.

However, we would like to leave one more comment: The true value of Galois theory lies more broadly in the connection between field extensions and subgroups of Galois groups, not as a tool for practically checking the solvability of specific polynomials. Especially, proving to the non-existence of generic quintic formula by Galois theory is like using a sledgehammer to crack a nut – it is way too powerful and complicated than needed for the problem at hand.

The following quote from Galois himself is a good reminder of the practical difficulty of applying Galois theory to specific polynomials³⁶

“If you now give me an equation that you have chosen at will, and you wish to know whether or not it is solvable by radicals, I will have nothing to do other than to indicate to you the way to respond to your question, without wishing to charge either myself or anyone else with doing it. In a word, the calculations are impractical. [...] But most of the time in applications [...] one is led to equations all of whose properties one knows beforehand: properties by means of which it will always be easy to answer the question by the rules which we shall expound.”

E. Galois [3] pp. 227, 229

In spite of the shortcomings (mostly on practical usage), Galois theory is an exemplary piece of mathematics in both theoretical and historical aspects. It is a groundbreaking work that has revolutionized the way we understand polynomials and their roots, and has laid the foundation for modern algebra.

Abel and Ruffini’s proof of the non-existence of generic quintic formulas terminated the era of algebra being primarily a subject of solving algebraic equations, while Galois’s work sets an example of solving a concrete problem by studying abstract algebraic structures, designating the future direction of algebra for the next two centuries.

We hope this drama has given you a glimpse of the beauty and power of abstract algebra, inspiring you to explore more of this fascinating field.

³⁶Determining the Galois group of a given polynomial can be rather tricky and infeasible; see this [\[post\]](#) on Math Stack Exchange for more info.

8 Bibliography

References

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A Appendix

A.1 Concepts in abstract algebra

Definition A.1 (Algebra). *In the history of mathematics, every **algebra** consists of a set and certain operations on that set. Here, an **operation** is simply a way of combining any two members of a set to produce a unique third member of the same set.*

It is usually said that the set is **closed** under the operation.

In modern notions, an **algebraic structure** is understood to be an arbitrary set, with one or more operations defined on it, and thus the subject of (abstract) algebra is *the study of algebraic structures*.

Example A.1 (Boolean algebra). *Let S be a set, and $\mathcal{P}(S)$ to be the set of all its subsets (called the **power set** of S). The set $\mathcal{P}(S)$, together with union $\cup$ and intersection $\cap$ as two operations defined on it, form the Boolean algebra, which is widely used in graphic design.*

A.2 Concepts in abstract algebra – Group and Abelian group

Definition A.2 (Group). *A **group** is a set G with an operation $*$ (called **group multiplication**, sometimes as $\cdot$ or simply omitted) which satisfies the group axioms:*

- *$*$ is associative: $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.*

*In this way, the expressions like $a * b * c * \dots * p$ is uniquely defined once the order of the elements involved are fixed.*

- *There is an element $e \in G$ such that $a * e = e * a = a$ for every $a \in G$.*

*The element e is unique [exercise] and called the **neutral element**.*

- *For every $a \in G$, there is an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.*

*The element a^{-1} is called the **inverse** of a . Every element a has a unique inverse [exercise].*

If the operation $*$ of a group is **commutative**, i.e., $a * b = b * a$ for every pair of $a, b \in G$, then we say this group is an **Abelian group** (named after Niels Henrik Abel).

Here are some examples of groups for you to get a taste of the concept:

Example A.2. *The set of integers $\mathbb{Z}$, with the operation of addition $+$, forms an Abelian group. The neutral element is 0, and the inverse of an integer a is $-a$.*

Example A.3. *The set of nonzero rationals $\mathbb{Q}^*$ with multiplication $\cdot$, forms an Abelian group. The neutral element is 1, and the inverse of $a \in \mathbb{Q}^*$ is $\frac{1}{a}$.*

This is not the only group structure on rational numbers: $\mathbb{Q}$ with addition $+$ also forms an Abelian group.

Example A.4. The set $\{0, 1, 2, 3, 4, 5\}$ with the operation of “addition modulo 6” forms an Abelian group commonly denoted by $\mathbb{Z}_6$. The neutral element is 0, and the inverse of a is $6 - a$. You can extend it to $\mathbb{Z}_n$ for any positive integer n .

A.3 Concepts in abstract algebra – Group of permutations

Consider that we have n objects, denoted by $1, 2, \dots, n$. A **permutation** of these n objects is a rearrangement of them, e.g., rearranging $(1, 2, 3, 4, 5)$ to $(3, 1, 5, 4, 2)$, commonly denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 5 & 4 & 2 \end{pmatrix},$$

which sends 1 to 3, 2 to 1, 3 to 5, 4 to 4, and 5 to 2.

It is instructive to regard a permutation as a *bijective* function from the set $\{1, 2, \dots, n\}$ to itself. For instance, the above permutation is a function f such that $f(1) = 3$, $f(2) = 1$, $f(3) = 5$, $f(4) = 4$, and $f(5) = 2$.

There are in total $n!$ distinct permutations of n objects, forming a set denoted by S_n .

Given two permutations f and g of n objects, we can compose them to get a new permutation $f \circ g$, which sends each object i to $f(g(i))$. (Function composition is by definition associative.) For example,

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 5 & 4 & 2 \end{pmatrix}, \quad g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix},$$

then $f \circ g$ sends $(1, 2, 3, 4, 5)$ to $(1, 5, 3, 2, 4)$. Clearly, $f \circ g \neq g \circ f$.

Definition A.3 (Symmetric group). The set S_n of all permutations of n objects, together with the operation of composition $\circ$, forms a group called the **symmetric group** on n objects, also denoted by S_n . The neutral element is the identity permutation ε sending every object to itself, and the inverse of a permutation f is its inverse function f^{-1} (every bijective function has a bijective inverse).

The symmetric groups are essential for understanding the non-existence of quintic formulas.

A.4 Concepts in abstract algebra – Subgroups

Now we turn to investigate relations between groups.

Definition A.4. Subgroup Let G be a group with operation $*$, and H a non-empty subset of G . Then H is a **subgroup** of G if H is closed with respect to $*$ and inverses, i.e., for every $a, b \in H$, $a * b \in H$ and $a^{-1} \in H$.

It follows that H itself is a group under the same operation $*$ [exercise].

Example A.5. The set of even integers $2\mathbb{Z}$ is a subgroup of the group of integers $\mathbb{Z}$ under addition. The set of positive integers $\mathbb{Z}^+$ is not a subgroup of $\mathbb{Z}$, since it is not closed under inverses: the inverse of a positive integer a is not in $\mathbb{Z}^+$.

Example A.6. The integers $\mathbb{Z}$ under addition is a subgroup of rationals $\mathbb{Q}$ under addition.

A.5 Concepts in abstract algebra – Group isomorphism

Definition A.5 (Group isomorphism). Let G and H be two groups. A **group isomorphism** from G to H is a bijective function $f : G \rightarrow H$ such that for every $a, b \in G$, $f(a * b) = f(a) * f(b)$, where the first $*$ is the operation of G and the second $*$ is that of H . Whenever such an isomorphism exists, we say G is **isomorphic** to H and denote this by $G \cong H$.

When treating elements of groups as abstract objects, $G \cong H$ means G and H are essentially the “same” group. Practically, it is often easier to recognize that two groups G and H are not isomorphic than to show they are.

Example A.7. The group of integers $\mathbb{Z}$ under addition is isomorphic to the group of even integers $2\mathbb{Z}$ under addition, since the function $f : \mathbb{Z} \rightarrow 2\mathbb{Z}$ defined by $f(a) = 2a$ is an isomorphism.

You see the “same” given by isomorphism is a bit counterintuitive.

Theorem A.1 (Cayley’s theorem^{P95–97}). Every group, not only the ones built upon finite/countable sets, is isomorphic to a group of permutations.

A.6 Concepts in abstract algebra – Cosets of group

In the previous subsection, we saw that a group ($\mathbb{Z}$) can interestingly be isomorphic to its subgroup ($2\mathbb{Z}$). In the rest (possibly tedious) discussions about groups, we will proceed to investigate more relationships between groups, especially between a group and its subgroups, as they are significant to understand Galois theory.

Definition A.6 (Cosets). Let G be a group and H a subgroup of G . For any $a \in G$, the symbol aH denotes the set $\{ah : h \in H\}$, and the symbol Ha denotes the set $\{ha : h \in H\}$. These two sets are called the **left coset** and **right coset** of H in G determined by a , respectively.

We will only be using right cosets, so we will simply call them **cosets**.

Theorem A.2. If $a \in Hb$, then $Ha = Hb$.

Proof. We prove this by showing $Ha \subseteq Hb$ and $Hb \subseteq Ha$.

Suppose $a \in Hb$. Then $a = h_1b$ for some $h_1 \in H$. For any $x \in Ha$, we have $x = h_2a$ for some $h_2 \in H$. Substituting $a = h_1b$, we get $x = h_2(h_1b) = (h_2h_1)b$. Since H is a subgroup, $h_2h_1 \in H$, so $x \in Hb$. Thus, $Ha \subseteq Hb$.

In the same way, for any $y \in Hb$, we have $y = h_3b$ for some $h_3 \in H$. Substituting $b = h_1^{-1}a$, we get $y = h_3(h_1^{-1}a) = (h_3h_1^{-1})a$. Since H is a subgroup, $h_3h_1^{-1} \in H$, so $y \in Ha$. Thus, $Hb \subseteq Ha$. $\square$

It follows that any two cosets Ha and Hb are either identical or disjoint: suppose $\exists x = Ha \cap Hb$, then $x = h_1a = h_2b$ for some $h_1, h_2 \in H$, so $a = h_1^{-1}h_2b \in Hb$, and therefore $Ha = Hb$ by the above theorem.

Theorem A.3. *The family of all cosets Ha , as a ranges over G , forms a **partition** of G . That is, every element of G belongs to exactly one coset of H .*

Proof. Since for every element $c \in G$, $c = ec$, where e is the neutral element of G , c belongs to at least one coset: Hc . Moreover, if c also belongs to another cosets Ha , then by the previous theorem, $Hc = Ha$. $\square$

It is instructive to point out that a coset may be written in multiple ways: if a coset of H contains n different elements $a_1, a_2, \dots, a_n$, it may be written in n different ways: $Ha_1 = Ha_2 = \dots = Ha_n$.

Theorem A.4. *If Ha is any coset of H , then there is a one-to-one correspondence between H and Ha .*

Proof. We prove this by showing the function $f : H \rightarrow Ha$ defined by $f(h) = ha$ is injective and surjective (a is fixed and h ranges over H).

f is injective: if $f(h_1) = f(h_2)$, then $h_1a = h_2a$, so $h_1 = h_2$.

f is surjective: every element in Ha has the form $ha = f(h)$. $\square$

By this theorem, if we consider finite groups, then every coset of H has the same number of elements as H . Together with the previous theorem, we can deduce the following **Lagrange's theorem**:

Theorem A.5 (Lagrange's theorem). *Let G be a finite group and H any subgroup of G . The number of elements in G (called the **order** of G) is a multiple of the order of H .*

A.7 Concepts in abstract algebra – Group homomorphism

Definition A.7 (Group homomorphism). *If G and H are two groups, a **group homomorphism** from G to H is a function $f : G \rightarrow H$ such that for any two elements $a, b \in G$, $f(a * b) = f(a) * f(b)$, where the first $*$ is the operation of G and the second $*$ is that of H . (An isomorphism is a bijective homomorphism.)*

If there exists a homomorphism from G **onto** (a surjective map) H , then H is called a **homomorphic image** of G .

Example A.8. The function $f : \mathbb{Z} \rightarrow \mathbb{Z}_6$ defined by $f(a) = a(\bmod 6)$ is a homomorphism from the group of integers under addition to the group of integers modulo 6 under addition. The homomorphic image of $\mathbb{Z}$ under f is $\mathbb{Z}_6$.

Passing from a group G to its homomorphic image H is like deliberately losing some aspects of G to get a simpler group H .

Theorem A.6. Let G and H be groups, and $f : G \rightarrow H$ a homomorphism. Then $f(e) = e$ (the e on the left (right) denotes the neutral element of G (H)) and $f(a^{-1}) = [f(a)]^{-1}$ for every $a \in G$.

Proof. As for any group, $a = a * e$, so $f(a) = f(a * e) = f(a) * f(e)$, which implies $f(e) = e$. Moreover, $f(a)f(a^{-1}) = f(a * a^{-1}) = f(e) = e$, so $f(a^{-1})$ is the inverse of $f(a)$. $\square$

A.8 Concepts in abstract algebra – Normal subgroup

Definition A.8 (Normal subgroup). Let H be a subgroup of G . If $\forall a \in H$ and $\forall x \in G$, $xa x^{-1} \in H$ (“closed with respect to conjugates”), then we call it a **normal subgroup** of G .

The notion of normal subgroup is crucial to understand a key structure in Galois theory: the quotient group, which will be discussed next.

A.9 Concepts in abstract algebra – Cosets of normal subgroup

Theorem A.7. If H is a normal subgroup of G , then $aH = Ha$ for every $a \in G$ (left and right cosets are equal).

Proof. For every $x \in aH$, we have $x = ah$ for some $h \in H$. Since H is normal, $a^{-1}ha \in H$, so $x = ah(a^{-1}a) = (aha^{-1})a = h'a \in Ha$, because $h' = aha^{-1} \in H$. Thus, $aH \subseteq Ha$. Analogously, one can show $Ha \subseteq aH$ and conclude $aH = Ha$. $\square$

This theorem gives a stronger reason that when studying the cosets of a normal subgroup, we don’t need to specify left or right.

Definition A.9 (Coset multiplication). There is a way of combining cosets, called **coset multiplication**:

$$Ha \cdot Hb = H(ab)$$

An immediate question about coset multiplication is whether it is well-defined: if $Ha = Hc$ and $Hb = Hd$, do we have $H(ab) = H(cd)$? The answer is no in general^{P148}, but yes if H is a normal subgroup of G :

Theorem A.8. Let H be a normal subgroup of G . If $Ha = Hc$ and $Hb = Hd$, then $H(ab) = H(cd)$.

Proof. Since $Ha = Hc$, $c \in Ha$ and thus $\exists h_1 \in H$ such that $c = h_1a$, similarly $\exists h_2 \in H$ such that $d = h_2b$. Since H is normal, $ah_2 \in aH = Ha$, so $ah_2 = h_3a$ for some $h_3 \in H$. Therefore, $cd = h_1ah_2b = h_1h_3ab$. Since $h_1h_3 \in H$, we have $cd \in H(ab)$. The by Theorem A.2, $H(ab) = H(cd)$. $\square$

A.10 Concepts in abstract algebra – Quotient groups

We are now ready to introduce the notion of quotient group. First, let H be a normal subgroup of G , denote G/H to be the set of all cosets of H

$$G/H = \{Ha : a \in G\},$$

and define an operation on G/H by $Ha \cdot Hb = H(ab)$, which gives us:

Definition A.10 (Quotient group). G/H with the coset multiplication is a group, called the **quotient group** of G by H .

We verify the group axioms:

- Associativity: $(Ha \cdot Hb) \cdot Hc = H(ab) \cdot Hc = H((ab)c)$, and $Ha \cdot (Hb \cdot Hc) = Ha \cdot H(bc) = H(a(bc))$. They are equal since $(ab)c = a(bc)$ in G .
- Neutral element: He is the neutral element of G/H , since $He \cdot Ha = H(ea) = Ha$ and $Ha \cdot He = H(ae) = Ha$.
- Inverse: $(Ha)^{-1} = H(a^{-1})$, since $Ha \cdot H(a^{-1}) = H(aa^{-1}) = He$ and $H(a^{-1}) \cdot Ha = H(a^{-1}a) = He$.

The quotient group inherits partial properties of the original group G :

Theorem A.9. G/H is a homomorphic image of G .

Proof. The most obvious function from G to G/H is the function f sending each element $a \in G$ to its own coset Ha : $f(a) = Ha$. It is a homomorphism: $f(ab) = H(ab) = Ha \cdot Hb = f(a) \cdot f(b)$. Moreover, f is surjective since for every $Ha \in G/H$, $f(a) = Ha$. $\square$

This theorem is applied when we prove the Galois's criterion of solvability in Sec. 5.15.

A.11 Concepts in abstract algebra – Abelian quotient groups

Theorem A.10. Let H be a subgroup of G , then (i) $Ha = Hb$ if and only if $ab^{-1} \in H$, and (ii) $Ha = H$ if and only if $a \in H$.

Proof. If $Ha = Hb$, then $a \in Hb$ and thus $\exists h \in H$ such that $a = hb$, so $ab^{-1} = h \in H$. Conversely, if $ab^{-1} \in H$, then $\exists h \in H$ such that $ab^{-1} = h$, so $a = hb \in Hb$, and thus $Ha = Hb$ by Theorem A.2. $\square$

We can derive an important fact at the bottom of Sec. 5.13 about the **commutator** $xyx^{-1}y^{-1}$ for two elements $x, y \in G$ and Abelian quotient groups: To say a quotient group G/H is Abelian is to say that $\forall x, y \in G$, $HxHy = HyHx$, so $Hxy = Hyx$, and thus $xy(yx)^{-1} = xyx^{-1}y^{-1} \in H$ by the above theorem. In short:

Theorem A.11. G/H is Abelian $\iff$ all the commutators $xyx^{-1}y^{-1} \in H$.

A.12 Concepts in abstract algebra – kernel and FHT

The last topic we want to introduce about group theory bears a huge title: **the fundamental homomorphism theorem (FHT)**. It is the key in proving the fundamental theorem of Galois theory.

Definition A.11 (Kernel of group homomorphism). *Let $f : G \rightarrow H$ be a group homomorphism. The **kernel** of f is the set of all elements in G that are sent to the neutral element of H :*

$$\text{Ker}(f) = \{x \in G : f(x) = e\}.$$

It is not hard to show that $\text{Ker}(f)$ is a normal subgroup of G [exercise]^{P141}.

Theorem A.12 (The fundamental homomorphism theorem (FHT)). *Let $f : G \rightarrow H$ be a homomorphism of G onto H , and $K = \text{Ker}(f)$. Then G/K is isomorphic to H .*

A.13 Concepts in abstract algebra – Proof of FHT

To prove the FHT let's first show a lemma:

Lemma A.13. *Let $f : G \rightarrow H$ be a homomorphism with kernel K . Then $f(a) = f(b)$ if and only if $Ka = Kb$, i.e., a and b belong to the same coset of K .*

Proof. $f(a) = f(b) \iff f(a)[f(b)]^{-1} = e \iff f(ab^{-1}) = e \iff ab^{-1} \in K$. By part (i) of Theorem A.10, $ab^{-1} \in K \iff Ka = Kb$. $\square$

This lemma says that all elements in a same coset of K have the same image under f , and conversely, elements which have the same image are in the same coset of K , giving rise to a one-to-one correspondence ϕ between the cosets of K and the elements of H : $\phi(Ka) = f(a)$, $\forall a \in G$ – a bijective map between G/K and H . Moreover, ϕ is an isomorphism: $\phi(Ka \cdot Kb) = \phi(K(ab)) = f(ab) = f(a)f(b) = \phi(Ka) \cdot \phi(Kb)$. Thus, G/K is isomorphic to H , completing the proof of FHT. $\square$

A.14 Concepts in abstract algebra – concluding remarks of group theory

This is the end of the first part of the appendix, which is about the basics of group theory. We have introduced the concepts of group, subgroup, isomorphism, homomorphism, normal subgroup, cosets, quotient groups, and the fundamental homomorphism theorem. These concepts are essential to understand Galois theory, as they will be used to establish the connection between field extensions and Galois groups.

The construction of the above discussions is purely based on the needs of understanding our drama of Galois theory, so many other important aspects of group theory are not covered. Moreover, we only provide shallow discussions on a limited amount of concrete examples³⁷. Readers who are interested in learning more about group theory are encouraged to read relevant textbooks.

³⁷The only somewhat carefully treated example are permutations and symmetric groups, as they are crucial in Galois theory.

A.15 Concepts in abstract algebra – Rings

Definition A.12 (Ring). A **ring** is a set A with operations called addition (+) and multiplication (symbolized as $\cdot$ or simply nothing: $ab = a \cdot b$) which satisfy the following axioms:

- A with addition alone is an Abelian group.
- Multiplication is associative: $a(bc) = (ab)c$ for all $a, b, c \in A$.
- Multiplication is distributive over addition. That is, $a(b+c) = ab+ac$ and $(b+c)a = ba+ca$ for all $a, b, c \in A$.

The neutral element for addition is called the **zero element**, usually denoted by 0. The additive inverse of an element a , called its **negative**, is usually denoted by $-a$.

When multiplication of a ring A is also commutative, then we say A is **commutative**. If there is in A a neutral element for multiplication, called **unity**, usually denoted by 1, then we say A is a **ring with unity**.

A.16 Concepts in abstract algebra – Integral domains

Definition A.13 (Integral domain). An **integral domain** is a commutative ring with unity (CRU) that has no zero divisors.

“No zero divisor” means if $ab = 0$ for $a, b \in A$, then either $a = 0$ or $b = 0$. This property is equivalent to [exercise] the **cancellation law**: from $ab = ac$ we can deduce $b = c$ whenever $a \neq 0$.

A.17 Concepts in abstract algebra – Fields

Definition A.14 (Field). A **field** is a commutative ring with unity (CRU) in which every nonzero element is invertible.

Every field is an integral domain. To prove this, let F be a field and $a, b \in F$ such that $ab = 0$. If $a \neq 0$, then a is invertible, so multiplying both sides of $ab = 0$ by a^{-1} gives $b = 0$. Thus, F has no zero divisors and is an integral domain.

On the other hand, not every integral domain is a field. For example, the set of integers $\mathbb{Z}$ is an integral domain but not a field since only 1 and -1 are invertible in $\mathbb{Z}$.

A.18 Proof: Product basis of vector spaces

Here we prove a fact about the product basis of vector spaces which we stated earlier:

Theorem A.14 (Product basis of vector spaces). Let $\{a_1, \dots, a_m\}$ be a basis of the vector space K over F , and let $\{b_1, \dots, b_n\}$ be a basis of the vector space E over K . Then the set of mn products $\{a_i b_j\}$ is a basis of the vector space E over the field F .

(The proof below is adapted from [1], pp. 294–295.)

Proof. Every element of E can be written as a linear combination of the $c = k_1b_1 + \cdots + k_nb_n$ with coefficients $k_i \in K$. But each k_i , being in K , can be written as a linear combination $k_i = l_{i1}a_1 + \cdots + l_{im}a_m$ with coefficients $l_{ij} \in F$. Substituting,

$$c = (l_{11}a_1 + \cdots + l_{1m}a_m)b_1 + \cdots + (l_{n1}a_1 + \cdots + l_{nm}a_m)b_n = \sum_{i=1}^m \sum_{j=1}^n l_{ij}(a_i b_j).$$

We still need to show that the mn products $a_i b_j$ are linearly independent over F : if $\sum_{ij} l_{ij}(a_i b_j) = 0$ if and only if all the coefficients $l_{ij} = 0$. Suppose

$$(l_{11}a_1 + \cdots + l_{1m}a_m)b_1 + \cdots + (l_{n1}a_1 + \cdots + l_{nm}a_m)b_n = 0,$$

and since $b_1, \dots, b_n$ are linearly independent, $l_{i1}a_1 + \cdots + l_{im}a_m = 0$ for every i . But $a_1, \dots, a_m$ are also linearly independent, so $l_{ij} = 0$ for every i, j . This concludes the proof of the product basis of vector spaces. $\square$

A.19 Proof: $\sigma(p + qr_1) = p + qr_2$ respects multiplication and division

For the product $(p + qr_1) \cdot (p' + q'r_1) = pp' + (pq' + p'q)r_1 + qq'r_1^2$, notice that $r_1^2 = -\frac{b}{a}r_1 - \frac{c}{a}$ by the original equation and hence the product is $(pp' - qq'\frac{c}{a}) + (pq' + p'q - qq'\frac{b}{a})r_1$. The mapping σ sends it to $(pp' - qq'\frac{c}{a}) + (pq' + p'q - qq'\frac{b}{a})r_2$. On the other hand, $\sigma(p + qr_1) \cdot \sigma(p' + q'r_1) = pp' + (pq' + p'q)r_2 + qq'r_2^2$, and again with $r_2^2 = -\frac{b}{a}r_2 - \frac{c}{a}$ it also becomes $(pp' - qq'\frac{c}{a}) + (pq' + p'q - qq'\frac{b}{a})r_2$. This concludes that $\sigma(x \cdot y) = \sigma(x) \cdot \sigma(y)$ for any $x, y \in K$.

You may also want to prove that σ respects division (multiplying inverse), which this is now trivial: By respecting multiplication, we know that $\sigma(x) \cdot \sigma(x^{-1}) = \sigma(x \cdot x^{-1}) = \sigma(1) = 1$, implying $\sigma(x^{-1}) = [\sigma(x)]^{-1}$. Therefore,

$$\sigma(x \cdot y^{-1}) = \sigma(x) \cdot \sigma(y^{-1}) = \sigma(x) \cdot [\sigma(y)]^{-1}$$

A.20 Proof: properties of field isomorphism

Let $\sigma : F_1 \rightarrow F_2$ be a field isomorphism, defined by $\sigma(x + y) = \sigma(x) + \sigma(y)$ and $\sigma(x \cdot y) = \sigma(x) \cdot \sigma(y)$ for every $x, y \in F_1$. We want to show that $\sigma(0) = 0$, $\sigma(1) = 1$, $\sigma(-x) = -\sigma(x)$, and $\sigma(x^{-1}) = [\sigma(x)]^{-1}$ for every $x \in F_1$.

- $\sigma(0) = \sigma(0 + 0) = \sigma(0) + \sigma(0)$, so $\sigma(0) = 0$.
- $\sigma(1) = \sigma(1 \cdot 1) = \sigma(1) \cdot \sigma(1)$, so $\sigma(1) = 1$.
- $\sigma(-x) = \sigma(0 - x) = \sigma(0) - \sigma(x) = -\sigma(x)$.
- $\sigma(x^{-1}) = [\sigma(x)]^{-1}$ already shown in the previous slide.